

Universität Hamburg
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Master Thesis

The impact of tropospheric versus stratospheric CO2 forcing on the Hadley cell

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Abstract

The Hadley cell (HC) is the meridional large scale circulation governing atmospheric flow in the tropics. Its ascending branch is associated with precipitation and the inter-tropical convergence zone (ITZC), while its descending branch coincides with very dry regions.

With climate change the Hadley circulation is expected to widen and weaken modulating precipitation patterns. Previous studies have shown that this can be mainly attributed to the increase in vertical stability of the atmosphere. Although CO2 is uniformly distributed in the atmosphere, in our experiments we have decomposed the CO2 forcing into a tropospheric and a stratospheric component using a control tropopause height climatology.

We found the response to be approximately linear with the stratospheric CO2 being responsible for a cooling in the stratosphere and the tropospheric CO2 showing the typical tropospheric warming signal. These temperature anomaly patterns also change the structure of the mean meridional circulation. Unlike general expectations, the HC strength is increasing with tropospheric CO2 while stratospheric CO2 causes a HC weakening.

The angular momentum budget does not provide a satisfactory explanation for the changes in the Hadley circulation. The two limits of angular momentum conservation and eddy driven regimes were not able to explain this behavior, mostly because the flow is never actually close to one of these limits and there is no theoretical solution for the intermediate regime. Instead, we found the energetic framework of the atmosphere to be more insightful. In this view, the HC acts as the primary mechanism by which energy is redistributed within the tropics, thereby reducing the top-of-atmosphere radiative energy imbalance. Its ability to transport energy poleward is controlled by the maximum mass stream function (MSF), the HC strength, as well as by the gross moist stability (GMS) of the atmosphere. Furthermore, the required energy transport by the HC can be changed by the net energy input into the atmosphere (forcing), the feedback response due to an increase in surface temperature, and the energy partitioning between the mean circulation and the eddies. We find that only by considering a combination of these factors, we can explain the contrasting HC responses.

This study contributes to a more comprehensive understanding of HC response to different patterns of forcing and provides a valuable test of existing theories of atmospheric circulation change.

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1 Introduction

The Hadley cell (HC) is the dominant meridional large-scale atmospheric circulation in the tropics. Its ascending branch at the equator is collocated with inter-tropical convergence zone (ITCZ), which is characterized by large amounts of precipitation. Accordingly, the descending branch of the HC is associated with the dry, semi-arid, and arid regions of subsidence in the subtropics. The equatorward flowing air in the lower branch is deflected westward by the Coriolis force, forming the so called trade wind. Equivalently, the anti-trades are formed by the poleward flowing air of the upper branch, which is deflected eastward. In the annual mean and in the equinoxes (see Figure 1.1) there are two similar overturning cells in the tropics - one in each hemisphere. The HC extends from 0 to 30 °N in the northern hemisphere and equivalently 0 to 30 °S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong cross-equatorial winter and a weaker summer HC (hemisphere with maximum insolation). This seasonal migration is related to the global monsoon, which has been identified as an angular momentum conserving HC regime (Bordoni & Schneider, 2008).

The HC is responsible for the poleward transport of energy and the equatorward transport of moisture, thereby reducing the meridional temperature gradients imposed by solar insolation. Thereby, the HC has strong implications for human and animal life as well as agriculture and vegetation on this planet. Its extent crucially modulates global precipitation patterns by setting the regions of ascend and descend. The strength of the HC determines the amount of energy is reaching the off-equatorial regions and the amount of moisture is reaching the equator, controlling the potential precipitation amounts. This is particularly evident in the Sahel region, which is located approximately 10 °N of the equator on the African continent. This semi-arid region receives most of its precipitation in the boreal summer when the HC and the ITCZ migrate northward. Thus, the rainfall in the Sahel is dynamically linked to the HC position. Changes in the HC also affect the livelihood of the local population and economies that remain heavily dependent on agricultural production.

That the HC can vary substantially has also been shown for past climates. Paleoclimate simulations of the HC display large changes in HC strength and extend during the last glacial maximum and the mid holocene due to changes in orbital forcing (D'Agostino et al., 2017). This is also supported by evidence from tropical paleoclimate proxy records of precipitation changes (McGee et al., 2014). This past evidence of strong HC changes combined with the

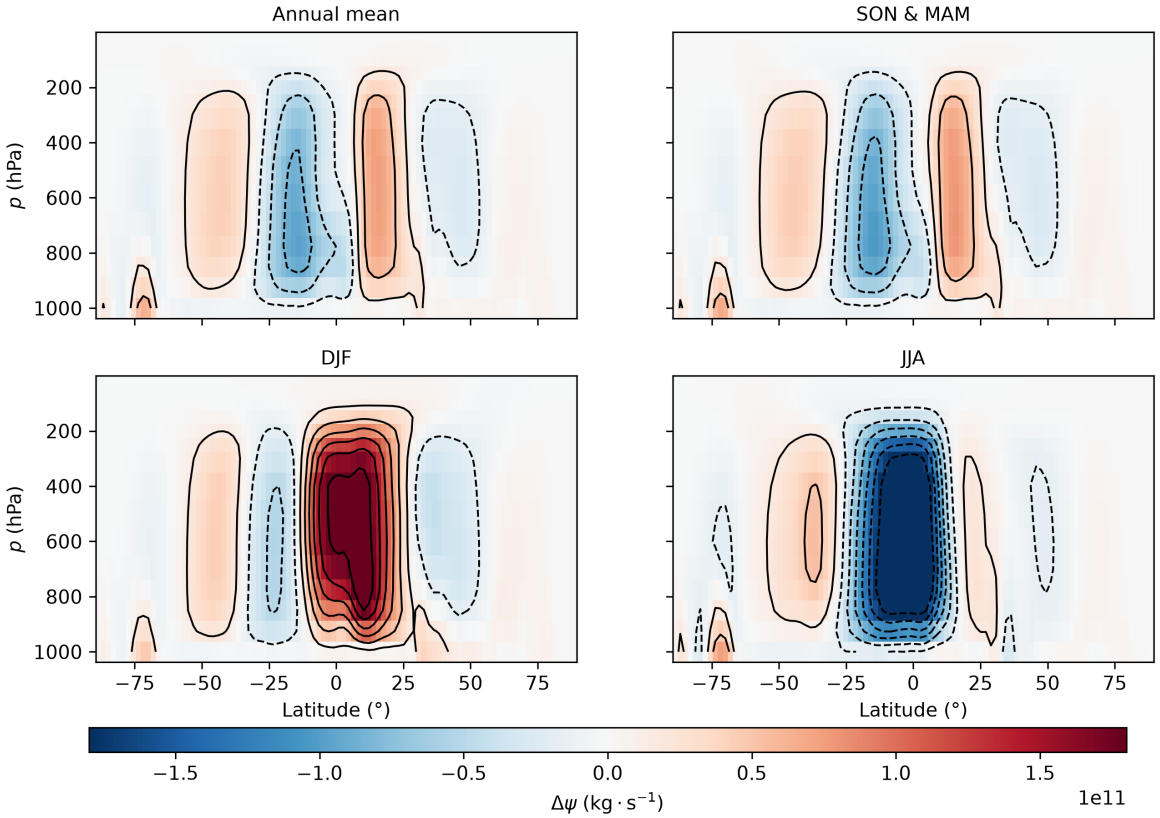


Figure 1.1: Annual mean and seasonal cycle of the meridional mass stream function in our PI control simulations (see Section 3.1). Both, the shading and the contours, indicate the mass stream flux. The two closest overturning circulations are the two HCs. Annual mean and equinox (SON&MAM) are very similar and bot approximately symmetric around the equator. In boreal winter (DJF), there is a strong winter and a weak summer HC. The same is true in southern hemisphere winter (JJA).

large societal impact of precipitation patterns give a strong reason to investigate the projected HC changes of human driven greenhouse gas emissions in the current era.

Climate simulations from the Coupled Model Intercomparison Project (CMIP), have established a HC weakening and widening under climate change (Lu et al., 2007; Xia et al., 2020). Explaining this, previous studies have emphasized the importance of drivers related to tropospheric temperature changes and profiles (Chemke & Polvani, 2021; Fan & Dommenget, 2024). However, historical simulations show that stratospheric forcing also influences the general circulation. In particular, stratospheric ozone depletion has led to a cooling of the Southern hemisphere lower-stratospheric polar cap. This has been identified as the main driver for the widening of the Hadley circulation over the second half of the twentieth century (Polvani et al., 2011). The goal of our study is to disentangle the contributions of tropospheric and stratospheric CO_2 to the established HC weakening and widening with a uniform CO_2 increase.

The CO_2 in the atmosphere is very uniformly distributed. The forcing that it imposes is not. The top of atmosphere forcing is critically set by the difference between surface and stratospheric temperature, which varies strongest in the meridional direction. Given that the

HC is flattening the meridional energy gradient, the response of the HC strength depends on meridional forcing structure. This has been shown for quadrupling of CO_2 only in certain latitude belts in coupled simulations (Kim et al., 2022). A forcing in mid latitudes weakens the Hadley circulation, whereas a forcing at the equator leads to it strengthening.

We investigate how the *vertical* structure of CO2 forcing, rather than its horizontal structure, alters atmospheric circulation. We achieve this by running simulations with a General Circulation Model (GCM), in which we quadruple CO_2 only in the troposphere and only in the stratosphere using a pre-industrial control tropopause height to separate the two cases. To our knowledge, these experiments have not yet been described in the literature. We find that the HC response to tropospheric CO_2 increase is a strengthening and widening in both hemispheres. This opposes our expectations evoked from previous studies, which identified tropospheric warming parameters as the main drivers for HC weakening under uniform CO_2 concentrations. Our results suggest that the weakening of the HC originates from stratospheric cooling caused by stratospheric CO_2.

The second part of this work investigates and explains the different responses of the HC strength, we employ angular momentum and energy budget frameworks. We use the axisymmetric angular momentum framework to attribute HC changes to changes in tropopause height, meridional temperature, gradient, and vertical stability. Since this theory ignores the contribution of eddies to HC strength, we also investigate changes in eddy divergence to explain the observed changes in the mean circulation. The energetic framework is a diagnostically unifying approach. It uses changes in top-of-atmosphere (TOA) radiative fluxes, eddy energy fluxes, ocean heat uptake, and moist static energy to quantify their contributions to the simulated HC changes. We will show that only a combination of all these factors can explain the opposing HC responses to tropospheric and stratospheric CO_2.

2

Theory

This chapter will provide the tools to understand the physical drivers behind HC changes. It also has the task to summarize the current state of knowledge regarding greenhouse gas forcing and the past as well as predicted Hadley cell changes which it might cause. Initially, we discuss how greenhouse gas forcing is changing the temperature structure and the circulation the atmosphere. Thereafter, we introduce the axisymmetric and eddy-driven HC dynamic framework. Combining the expected temperature structure changes, we are trying to develop a framework to understand the HC response to our non-uniform CO_2 forcing experiments. Finally, we introduce the energy budget decomposition framework, which will help in the analysis of the relative contributions of changes in the energy balance to changes in the Hadley circulation.

2.1. CO2 forcing in the atmosphere and its temperature response

CO_2 is relatively well mixed in the atmosphere. In our simulations with a general circulation model though, we can prescribe different vertical CO_2 profiles. Since CO_2 is a greenhouse gas, this changes the radiative balance and thus the temperature profile of the atmosphere. This section describes the temperature response of the atmosphere to a uniform increase in CO_2 concentration and establishes expectations for the temperature change in the different vertical CO_2 profiles. In particular, this sections cares about the tropopause height, and meridional and vertical temperature gradients, which are important to understand changes in the HC strength and extent as we will show in Section 2.3.1.

2.1.1. Tropospheric warming and stratospheric cooling

The temperature response to a uniform increase in CO_2 concentration is characterized by a warming in the troposphere and a cooling in the stratosphere. The vertical temperature profile of the atmosphere can be determined as a first approximation from the budget of incoming and outgoing radiation everywhere. This state is called radiative equilibrium (RE). However, in RE the temperature in the atmosphere rapidly decreases with height, rendering the atmosphere statically unstable. In reality, convection transports additional energy upward and relaxes the temperature profile to a moist adiabat, with a lapse rate of about $6.5\,\mathrm{K\,km^{-1}}$. The resulting atmospheric state is called to be in radiative-convective equilibrium (RCE)

(Manabe & Strickler, 1964). Thus, energy balance in the atmosphere is not just achieved by radiative balances but heavily influenced by convective redistribution of heat. Increasing the CO_2 concentration in the atmosphere raises the effective emission height in the CO_2 band from the surface to the stratosphere. Less energy leaves the atmosphere and stays trapped in the troposphere. This increases the temperature, thus the emission of longwave radiation, in the atmosphere until it is again in radiative balance. On the vertical the temperature change is redistributed via convection (Jeevanjee, 2023).

The temperature in the stratosphere is governed by the ratio of heating due to the absorption of shortwave radiation by ozone and the emission of longwave radiation due to CO_2, which cools the atmosphere. One of the most robust responses of increasing the CO_2 concentration in the atmosphere has been the cooling of the stratosphere (Santer et al., 1996). Because the heating due to ozone stays constant but the emissivity of CO_2 is increased, the atmosphere has to cool locally to stay in radiative balance. This is independent of surface temperature change which happens on annual to milennial time scales. Therefore, the stratosphere adjusts on the radiative time scale on the order of weeks to months. Near the tropopause changes in stratospheric temperature are almost negligible, while they increase with height until they reach their maximum approximately at the stratopause. Here, each doubling in CO_2 decreases the temperature by around 8 K. More insights into the structure of stratospheric cooling can be obtained from a spectroscopic analysis of CO_2 forcing in the atmosphere (Cohen et al., 2026).

An rise in tropopause height due to a uniform increase in CO_2 is robust at all latitudes in most models. A simple model for a grey atmosphere predicts that the HC changes either due to changes in surface temperature or in lapse rate. Increasing the surface temperature and decreasing the lapse rate pushes the tropopause up (Vallis et al., 2015). This is what happens under uniform CO_2 forcing. We would expect the same response for a tropospheric CO_2 increase. For a stratospheric CO_2 increase, we would expect no response in the tropopause height, since surface temperature and the lapse rate are not affected by stratospheric cooling.

Separating the CO_2 concentration change into a tropospheric and a stratospheric component will have a very different response in terms of the temperature anomaly field. For an increase of tropospheric CO_2 concentration, we expect a warming signal from the troposphere due to the increased absorption of CO_2. For an increase of stratospheric CO_2 concentration, we expect a cooling signal from the stratosphere due to the increased emission of radiation at these locations.

2.1.2. Warming structure and feedbacks

The warming and cooling pattern in the atmosphere due to a CO_2 increase does not yet reveal how vertical and meridional temperature gradients change. In the tropics, the warming aloft is amplified (top heavy). This is due to increased moisture content, which reduces the lapse rate. This constitutes a negative feedback. Due to the greenhouse gases a substantial fraction of outgoing longwave radiation (OLR) originates in the mid troposphere, where now the warming is amplified, which increases the OLR. Another important paradigm to explain the horizontal profiles of temperature within the tropics is the weak temperature gradient approximation (Sobel et al., 2001). It argues that the Coriolis force is too weak in the tropics to sustain horizontal gradients of pressure or temperature. Diabatic heat fluxes are communicated throughout the tropics by gravity waves maintaining the horizontal

homogeneity. These two concepts combined provide a good framework to understand the warming structure with an increase in CO_2 concentration in the tropical troposphere: The vertical temperature gradient is reduced which means that the atmosphere is more stable. Vertical temperature gradients within the tropics are expected to remain equal due to to the weak temperature gradient approximation.

In the Arctic, we have the somewhat opposite pattern. Warming is amplified at the surface (bottom heavy), and generally larger than at the rest of the globe. This phenomenon is called Arctic amplification. Enhanced warming is neither yet expected nor observed in the Antarctic, where it is delayed by enhanced ocean heat uptake (Armour et al., 2016) and the large elevation of Antarctic continental land mass (Hahn et al., 2020). Warming in the Arctic is enhanced during the winter months not during the summer as one would expect from the ice albedo feedback. The increased sea ice melt due to increased surface temperatures during summer increases the OLR and heat fluxes from the newly opened waters, further warming the atmosphere (Dai et al., 2019). This local effect is enhanced by remote influence on the local lapse rate. Climatologically, there is a surface-based inversion because advective moist heating keeps the atmosphere warm compared to the surface which strongly cools to space. The warming associated with the sea ice loss weakens the inversion, producing a positive lower lapse rate feedback. Generally, the atmosphere in the Arctic is quite dry and due to inversion convection cannot transport moisture into the free troposphere. This diminishes the upper lapse rate feedback in the Arctic decreasing the OLR relative to lower latitudes, further warming the Arctic (Feldl et al., 2020). The enhanced warming at lower level in the Arctic compared to the global mean, increases the meridional near-surface temperate gradient.

2.2. Historical and projected Hadley cell trends

The HC response to a uniform increase in CO_2 concentration and the associated temperature response is very well described in the literature. HC changes have been observed already since the beginning of the satellite era. Observational datasets using outgoing longwave radiation, precipitation, and sea level pressure as proxies for the HC position attest a significant widening of the HC in the historical period (1979-2000s) (Hu et al., 2011). Depending on the observed quantity, the widening had a magnitude of 1.2 to 4.0°. Similar trends have also been found in reanalysis data (Hu & Fu, 2007). In phase 6 of the Coupled Model Intercomparison Project (CMIP6) "All-forcing" experiments also found both Hadley cells widening during the period 1970-2014 by in mean $0.065° \pm 0.02°$ per decade. This is much smaller than the trends obtained from observational records. The influence of natural variability is highlighted in this context. The widening in CMIP6 models is found to be larger in the southern hemisphere. The primary driver of the historical widening was greenhouse gas forcing. Changes in anthropogenic aerosol forcing and stratospheric ozone only contributed to southern hemisphere HC changes (Xia et al., 2020).

In terms of the HC strength, studies found opposing trends in climate models and reanalysis data in the historical period for the northern hemisphere. While CMIP5 models show a robust weakening of the NH HC, the reanalysis data even suggests a strengthening. This discrepancy is not due to internal variability but rather stems from a misrepresentation of latent heating in the reanalysis (Chemke & Polvani, 2019). According to CMIP6 models, greenhouse gas forcing led to a weakening in both hemispheres, with stratospheric ozone,

and aerosol forcing contributing to a net modest strengthening in the southern hemisphere. This strengthening trend had not been observed in the previous CMIP5 models. The annual mean northern hemisphere HC weakened by approximately 5% over 1970-2014 (Xia et al., 2020).

Similar trends can be seen for future HC trends. CMIP6 projections of the HC extent show that the widening trend of the HC increases with a stronger radiative forcing in both hemispheres. This increase is stronger in the annual mean southern hemisphere HC and does not exhibit any seasonality. The strength of the SH HC is not projected to change with radiative forcing. In the NH, there is a projected net weakening of the annual-mean maximum of about 10% over the period of 2016-2100 (Xia et al., 2020). Consistent with the historical simulation the largest contributor to the change is expected to be the increase CO_2 concentrations in the atmosphere.

2.3. Dynamical theories of the Hadley circulation

Given the changes in the temperature profiles caused by the tropospheric, or stratospheric CO_2 forcing, we want to discuss their implications on the Hadley circulation. Thus, we investigate what HC changes are to be expected according to different HC frameworks. The conservation of angular momentum is the physical concept to understand the movements of fluids, the circulation of air, in the atmosphere imposed by a thermal structure. In the following subsections, we will introduce two different dynamical theories of the HC and discuss their respective predictions for HC strength and extent changes in respective to the changes in temperature.

2.3.1. Axisymmetric theory

Trying to answer the question why there is a meridional circulation in the first place, Held and Hou (1980) developed the axisymmetric theory of the HC. This question is raised because one could imagine a world where the Earth is just in radiative equilibrium (see Section 2.1.1) everywhere. This means that every point on Earth emits just as much radiation as it receives from the sun. Thus, there would not be any meridional heat flux necessary.

For the derivation of the circulation in the tropics, the axisymmetric theory considers the zonal angular momentum equation in the ϕ–p plane with no meridional circulation ($v = \omega = 0$). Being zonally symmetric prohibits the existence of large scale barotropic or baroclinic eddies. Choosing a radiative-equilibrium temperature profile, which resembles that of the Earth,

$$\frac{\theta_E}{\theta_0} = 1 - \frac{1}{3}\Delta_h(3\sin^2\phi - 1) + \Delta_v\left(\frac{z}{H} - \frac{1}{2}\right) \tag{2.1}$$

(see Figure 2.1a), one can obtain the corresponding zonal wind u_E via the thermal wind relationship. Here, Δ_h is the fractional pole to equator temperature difference, Δ_v the fractional temperature difference between the surface and the top of the fluid, H the depth of the fluid, and θ_0 the global mean value of θ_E. This solution does not only produce westerly winds at the equator, which are not observed in the atmosphere, but also violate Hide's theorem (Hide, 1969). Hide's theorem states that a maximum of angular momentum within the flow domain cannot be sustained due to dissipation within the flow. A maximum of angular

momentum can only exist at the boundaries. This tells us that this RE temperature profile cannot persist on Earth and that we require some poleward transport of heat to flatten the meridional temperature gradient.

The conservation of angular momentum

$$M = (u + \Omega a \cos\phi) a \cos\phi \tag{2.2}$$

provides a second constraint for a HC solution with the zonal wind u, a being the radius and Ω the rotational frequency of the Earth. The angular momentum conserving wind, which assumes that the zonal wind is zero at the equator, can then be derived as

$$u_M = \Omega a \frac{\sin^2\phi}{\cos\phi}. \tag{2.3}$$

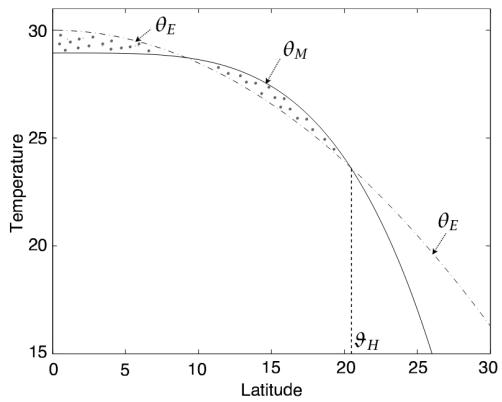
This allows to infer, via the thermal wind relationship, a temperature profile θ_M (see Figure 2.1a), which is now tied to angular momentum conservation. Assuming that the energy is conserved in the upper branch and that the solution is continuous at the edge of the HC, allows to determine the undefined parameters of this solution, one of which is the latitude of the HC edge

$$\phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 a^2}\right)^{1/2}. \tag{2.4}$$

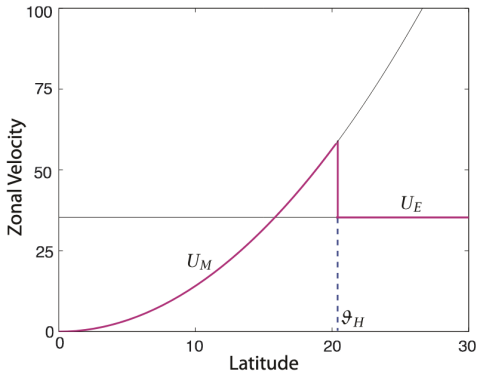
From the temperature equation, one also obtains a scaling for its strength

$$\psi_{\max} \sim \frac{1}{\tau}\frac{g^{3/2}(H\Delta_h)^{5/2}}{\Delta_v a^2 \Omega^3}. \tag{2.5}$$

Here, g is the gravitational acceleration of Earth and τ is the time scale at which θ relaxes to θ_E.



(a) The radiative-equilibrium temperature profile θ_E is flattened to the angular-momentum-conserving temperature θ_M by the HC.



(b) The angular-momentum-conserving zonal wind u_M increases with latitude. Poleward of the HC edge, the zonal wind from the radiative-equilibrium solution u_E is valid.

Figure 2.1: Latitudinal profiles of temperature and zonal wind under radiative equilibrium and under angular momentum conservation of the axisymmetric model, from Vallis (2017).

The key contribution of this theory is to explain why there is a HC with finite width in the first place. It is to reduce the meridional temperature gradients imposed by solar insolation.

For a uniform increase in CO_2 we know that the tropopause height increases, which would lead to a HC widening and strengthening according to Equations (2.4) and (2.5). The vertical temperature gradient of potential temperature increases (more stable atmosphere), which would lead to a weakening of the HC according to Equation (2.5). The meridional temperature gradient is expected to decrease at lower levels in the atmosphere due to arctic amplification. In contrast, it is expected to increase aloft due to stratospheric cooling on the poleward flank of the subtropical jet and the tropospheric warming at the equatorward flank. Thus, the axisymmetric theory cannot be used to qualitatively predict the HC response to a uniform increase in CO_2 in the atmosphere.

For an increase of tropospheric CO_2, we expect a warming of the troposphere, which is enhanced aloft and amplified in the arctic. This will increase the tropopause height, decrease the vertical temperature gradient, and decrease the meridional temperature gradient at lower levels while increasing it aloft. This reflects the same situation as for a uniform increase in CO_2, which is why we expect a weakening and a widening of the HC (as is evident in the uniform case). Just from the scaling laws of (2.4) and (2.5), qualitative change of the HC with an increase of tropospheric CO_2 cannot be predicted.

For an increase of stratospheric CO_2, we expect a cooling of the stratosphere. In the axisymmetric framework this would only affect the meridional temperature gradient, which is increased at the location of the subtropical jet. From this, we would expect a modest widening and strengthening from the axisymmetric scalings if any effect at all.

The axisymmetric theory predicts that the HC depends directly on thermal driving via the scaling laws. Subsequent studies have advanced this theory also to temperature profiles which have their maximum off the equator, in order to account for seasonal variations (Lindzen & Hou, 1988). In the axisymmetric model, the HC is weaker in equinox conditions than observed, leading to the suggestion that the HC in the annual mean is not close to equinox conditions but rather that the HC flips between solstitial states. More recent studies have addressed this discrepancy using eddy permitting models (Walker & Schneider, 2005). In these models, the strength of an annually averaged Hadley circulation is comparable to the Hadley circulation driven by an annually averaged heating. This means that the HC in its equinox state resembles very much an annual mean HC. This highlights the importance of eddies in driving the HC strength, which will be covered in the next section.

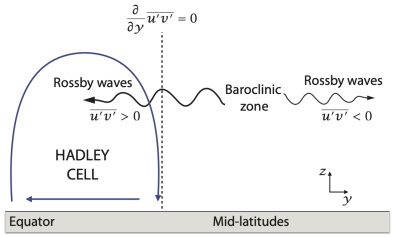


Figure 2.2: Schematic of the eddy driven HC, from (Vallis, 2017). The eddy momentum fluxes generated in the baroclinic zone transport momentum northward, inducing a mean circulation.

2.3.2. Eddy driven mean circulation

The importance of eddies in shaping the HC was acknowledged even before the axisymmetric theory was developed (Kuo, 1956). More recent studies have compared axisymmetric models

to eddy permitting simulations to determine the importance of eddies in altering HC strength (Singh et al., 2017; Walker & Schneider, 2006). They have found that the eddies increase the HC strength by 50–75%, depending on the model.

From the eddy perspective the HC edge is set by the latitude at which the zonal winds become baroclinically unstable. Baroclinic instability arises due to large vertical shear in a flow. This is the source of turbulent eddies in the flow field. Using a two layer model, one can derive the HC edge as (Held, 2000)

$$\phi_H \sim \left(\frac{NH_e}{\Omega a}\right)^{1/2} \tag{2.6}$$

with N being the Brunt-Väisälä frequency, and H_e the depth of the troposphere in the extratropical regions. It is important to note that this is a different scaling than the one derived from the axisymmetric theory. The eddy perspective predicts that the HC edge is set by extratropical parameters, whereas the axisymmetric theory predicts that it is set by tropical parameters.

To quantify HC strength changes from the eddy perspective, a Reynolds decomposition on the zonal and meridional wind is applied. One obtains the eddies, marked with a prime, as deviations in time from the zonal and temporal mean flow, marked by an overline,

$$u = \bar{u} + u', \quad v = \bar{v} + v'. \tag{2.7}$$

These deviations are associated with weather and Rossby waves in the mid-latitudes. Inserting this into the zonal momentum equation and neglecting viscous forces and vertical advection, gives

$$f\bar{v}(1 - Ro) = \frac{1}{a\cos^2\phi}\frac{\partial(\cos^2\phi\ \overline{u'v'})}{\partial\phi} + \frac{\partial\overline{u'\omega'}}{\partial p}. \tag{2.8}$$

Here, the Rossby number is

$$Ro = \frac{\bar{\xi}}{f} \tag{2.9}$$

where $\xi = \hat{k} \cdot \nabla \times \vec{u}$ is the relative vorticity. Equation (2.8) has two limits. For $Ro \to 1$, the left-hand side becomes zero and the eddy term becomes negligible. This means that the Hadley circulation is close to angular momentum conservation

$$\frac{\partial\overline{M}}{\partial\phi} = 0 \tag{2.10}$$

as described by the axisymmetric solution in Section 2.3.1. For $Ro \to 0$, the meridional wind $\bar{v}$ is instead balanced by the eddy momentum flux convergence. This regime can be understood as eddy-driven (see Figure 2.2). For the intermediate Ro, no complete dynamical theory exists (Schneider, 2006).

In the eddy driven limit, the climatological divergence of eddy momentum flux, which is observed in the tropics, requires a positive mean meridional wind. This holds even if the zones of baroclinic instability lie well poleward of the HC edge. Rossby waves can propagate into the HC, producing a poleward momentum flux. At the HC edge, there is

$\bar{v} = 0$ and thus no eddy momentum flux. The eddy momentum flux divergence also explains the equatorward meridional wind in the eddy-driven Ferrel cell in mid-latitudes. Over the seasonal cycle, the Rossby number varies between 0.2 and 0.8, approaching either of the two limits. During the equinoxes the Rossby number is small, which indicates a strong eddy influence. During the solstices the Rossby number is closer to 1 in the winter HC but still small in the summer HC.

Assuming that the HC is entirely eddy driven ($Ro = 0$), the HC strength can be predicted from the eddy momentum flux by inserting Equation (2.8) into Equation (3.1). Following Walker and Schneider (2006), this gives the integrated eddy momentum flux divergence

$$\psi_{\max} = \frac{2\pi a \cos\phi}{fg} \int_{p_{\max}}^{0} \left(\frac{1}{a\cos^2\phi} \frac{\partial(\cos^2\phi\ \overline{u'v'})}{\partial\phi} + \frac{\partial\overline{u'\omega'}}{\partial p} \right) \mathrm{d}p' \equiv \mathcal{T} \tag{2.11}$$

with $p_{\max}$ being the pressure level at which the MSF becomes maximum, and the Coriolis parameter f. The quantity $\mathcal{T}$ is a predictor for the HC strength given a certain eddy momentum flux divergence. The HC is thus only indirectly coupled to thermodynamic drivers, which set the zones of baroclinic instability. It has been shown to be skillfully predict the HC strength in eddy permitting models for various climates (Walker & Schneider, 2006).

From the eddy framework, one can predict how the temperature response to CO_2 forcing influences the HC. Therefore, it is crucial to look at how the zones of baroclinic instability change. This is set by the changes in meridional temperature gradient via the thermal wind relationship. The decrease in meridional temperature gradient at the surface opposes the effect of an increase in meridional temperature gradient aloft in both the uniform and the tropospheric CO_2 increase case. Which effect dominates the jet change, is still discussed in the literature (Ossó et al., 2024).

2.4. The energy budget framework of the Hadley circulation

The energy budget analysis provides a diagnostic perspective on the HC. It cannot explain why the Earth is not in radiative (convective) equilibrium, but provides a framework to understand its energy transport mechanisms. Thus, it circumvents the Rossby number regime dependency, providing a more universal approach. Important to notice is that this framework can only deliver a prognostic relationship between energy transport and HC strength. Changes in HC extent cannot be investigated.

Using TOA radiative fluxes (see Figure 2.3), one can infer that the short-wave radiative flux is much larger than the emitted outgoing long wave radiation at the equator . The opposite is true for the poles. This implies a poleward energy transport by the atmosphere and the ocean. For every column of the earth system, the energy tendency can be written as

$$\frac{\partial E_{\mathrm{ao}}}{\partial t} = R_{\mathrm{TOA}} - \Delta F_{\mathrm{ao}} \tag{2.12}$$

where R_{TOA} is the net radiative flux at the top of the atmosphere, which consists of the net shortwave radiation plus the net longwave radiation (positive values indicate a downward flux), and ΔF_{ao} is the divergence of energy transport by the atmosphere and ocean (Hartmann, 2016; Oort & Haar, 1976).

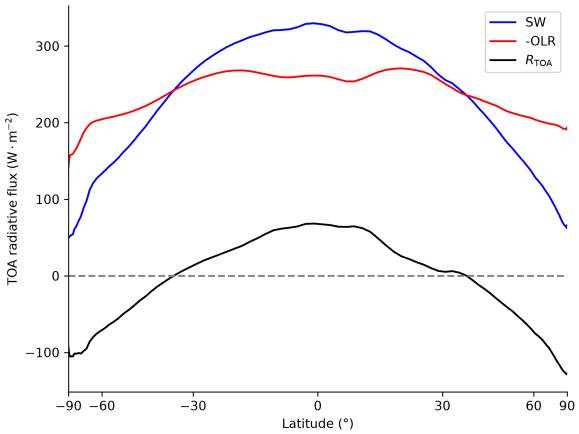


Figure 2.3: Zonal and annual mean average of net shortwave and net outgoing longwave radiation at the TOA from PI control (see Section 3.1) simulations. The sum shows a net surplus of energy in the tropics and an energy deficiency in the extra-tropics, implying a poleward energy transport.

Integrating this equation in the steady state until latitude ϕ gives

$$F_{ao}(\phi) = \int_{-\pi/2}^{\phi} \int_{0}^{2\pi} R'_{\mathrm{TOA}} a^2 \cos\phi \mathrm{d}\lambda \mathrm{d}\phi', \tag{2.13}$$

which provides a measure for the energy that has to pass through this latitude. One should use the anomalous net radiative TOA flux R'_{TOA} here, subtracting the global mean value, to make sure that the energy transport is closed (zero transport at both poles) (Zelinka & Hartmann, 2012). To separate between the energy transport by the atmosphere F_a and the ocean, one can subtract the surface energy fluxes, which can be obtained from the sum of surface latent, sensible heat, and net shortwave and longwave fluxes, from the TOA radiative fluxes.

To understand how the changes in the energy transport are related to changes in the HC, we need to quantify the energy transport by the HC. For this, we need to determine the energy within an air parcel in the atmosphere, which is given by the moist static energy (MSE). It is defined as

$$h = c_p T + gz + L_v q \tag{2.14}$$

with c_p being the specific heat capacity at constant pressure, g the gravitational acceleration, z the height, L_v the latent heat of vaporization and q the specific humidity. The transport of MSE by the mean meridional circulation at a certain latitude ϕ is given by the vertical integration of the adjusted meridional wind $\bar{v}_{\mathrm{adj}}$ multiplied by the MSE h, which is the meridional energy flux, over the entire atmospheric column

$$F_{\mathrm{HC}}(\phi) = \frac{2\pi a \cos\phi}{g} \int_{0}^{p_s} [\bar{v}_{\mathrm{adj}}][\bar{h}] \mathrm{d}p. \tag{2.15}$$

This expression compares the poleward energy transport by the upper branch of the HC to the equatorward energy transport by the lower branch. The adjusted meridional wind is defined as

$$[\bar{v}_{\mathrm{adj}}] = [\bar{v}_+] - [\bar{v}_-] \frac{\{[\bar{v}_+]\}}{\{[\bar{v}_-]\}}, \tag{2.16}$$

where $[\bar{v}_+]$ denotes northward zonal mean wind (i.e., equal to $[\bar{v}]$ if $[\bar{v}] > 0$ and zero elsewhere), and $[\bar{v}_-]$ defined conversely with $\{\cdot\} = \int \frac{\cdot}{g} \mathrm{d}p$ indicating a vertical mass weighted integral over the depth of the column. This transformation guarantees that the meridional mass flow is strictly in mass balance with $\{[\bar{v}_{\mathrm{adj}}]\} = 0$. It provides a correction for imbalances produced by the interpolation from sigma to pressure vertical levels. Figure 2.4 shows the energy transport by the mean circulation, which is largest in the tropics where it is called the Hadley cell.

Expressing the HC strength $\psi_{\max}$ in terms of the energy transport F_{HC} and the gross moist stability $\mathcal{H}$ gives

$$F_{\mathrm{HC}} = \psi_{\max} \mathcal{H} \tag{2.17}$$

The gross moist stability (GMS) is defined as the ratio of energy transport to mass transport by the HC which can be understood as the energy transport efficiency. It is inferred to

$$\mathcal{H}(\phi) = \frac{2\pi a \cos\phi}{g} \frac{1}{\psi_{\max}} \int_0^{p_s} [\bar{v}_{\mathrm{adj}}][\bar{h}] \mathrm{d}p. \tag{2.18}$$

GMS is a measure for the stratification of the atmosphere. A large GMS indicates a strongly stratified atmosphere. To understand the changes in HC strength, we linearize Equation (2.17) and obtain

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\mathrm{HC}}}{F_{\mathrm{HC}}} - \frac{\delta\mathcal{H}}{\mathcal{H}}. \tag{2.19}$$

This yields the contribution of GMS and HC transport changes. Knowing the total energy transport in the atmosphere F_{a} from Equation (2.13) (subtracting surface fluxes), it can be further split up into a mean F_{HC} and an eddy energy flux component F_{e} (see Figure 2.4), which gives

$$F_{\mathrm{a}} = F_{\mathrm{HC}} + F_{\mathrm{e}}. \tag{2.20}$$

The eddy component can be calculated as the difference between F_{a} and F_{HC} or by using the transient eddy fluxes of MSE

$$F_{\mathrm{e}}(\phi) = \frac{2\pi a \cos\phi}{g} \int_0^{p_s} [\overline{v'h'}] \mathrm{d}p'. \tag{2.21}$$

Inserting this into the linearized Equation (2.15) gives

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\mathrm{a}}}{F_{\mathrm{HC}}} - \frac{\delta F_{\mathrm{e}}}{F_{\mathrm{HC}}} - \frac{\delta\mathcal{H}}{\mathcal{H}}.$$

The transport by the atmosphere can be expressed as

$$F_{\mathrm{a}} = F_{\mathrm{ao}} - F_{\mathrm{o}} \tag{2.22}$$

with F_{o} being the transport due to ocean heat uptake (see Figure 2.4). The HC changes are now a function of the total change in energy transport.

The changes in R_{TOA} can be further split up into a change in radiative forcing R_f and a change due to feedbacks $\sum_i \lambda_i \Delta T_s$, which gives

$$\Delta R_{\mathrm{TOA}} = \sum_i \lambda_i \Delta T_s + R_f. \tag{2.23}$$

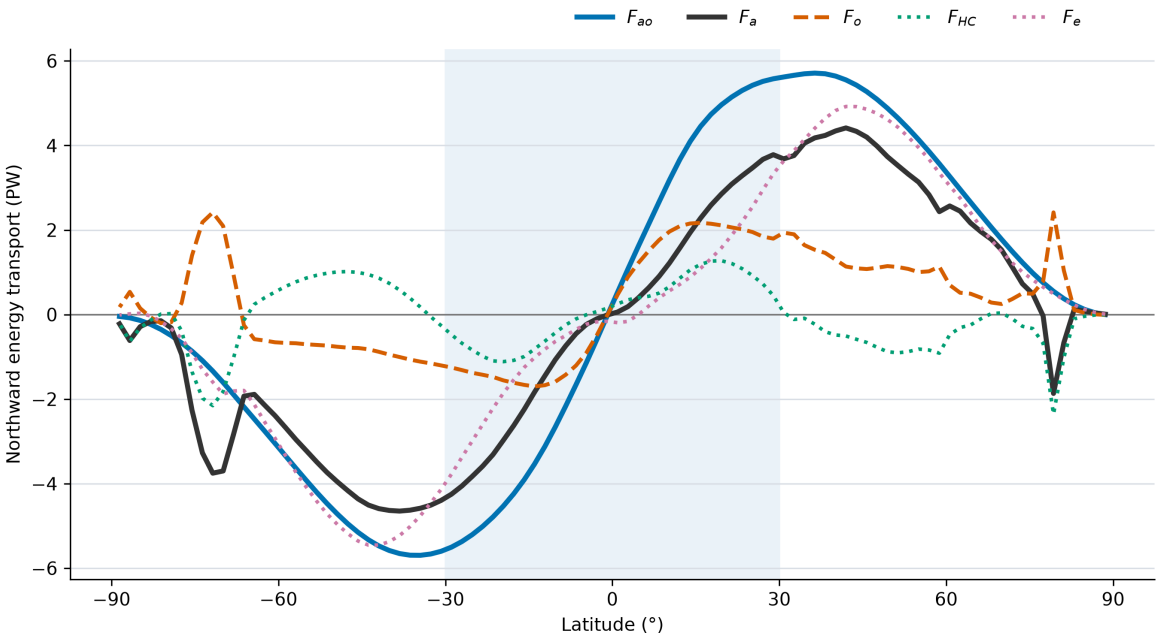


Figure 2.4: Northward energy transport by the global Earth system (F_{ao}), by the atmosphere(F_a), by the ocean (F_o), by the mean circulation (F_{HC}), and the eddies (F_e) the PI control simulation (see Section 3.1). The tropics are highlighted in light blue.

The forcing is the temperature-independent effect of CO2 on the TOA radiative fluxes and the feedbacks are the changes in TOA radiative fluxes due to changes in surface temperature. This change in surface temperature has an influence on the surface albedo, water vapor, clouds, and lapse rate, and OLR. If the increase of surface temperatures increases or decreases the downward radiation in relation to a specific physical mechanism, it is called a positive or negative feedback, respectively.

Inserting Equation (2.23) into Equation (2.13) gives us the final expression for the HC strength changes decomposed into the contributions of forcing δF_{forc}, feedbacks δF_{fdb}, ocean heat uptake F_{o}, eddies δF_{e}, and GMS δH

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\text{forc}}}{F_{\text{HC}}} + \frac{\delta F_{\text{fdb}}}{F_{\text{HC}}} - \frac{\delta F_{\text{o}}}{F_{\text{HC}}} - \frac{\delta F_{\text{e}}}{F_{\text{HC}}} - \frac{\delta H}{H}. \tag{2.24}$$

GMS is expected to increase with global warming, which is, according to this framework, contributing to a weakening of the HC. Again, using the known changes of these parameters under uniform CO_2 forcing, we are trying to qualitatively predict their contribution to the HC changes in our respective forcing cases. A uniform increase in CO_2 increases the GMS, which contributes to a weakening of the HC. An increase in eddy heat transport also contributes to a decreases in HC strength, because more of the necessary transport carried out by the eddies. In our particular setup (see Section 3.1), OHU is prescribed by surface energy fluxes that mimic the ocean transport. We will later see, that this is not exactly true in our setup. In CMIP models, studies find a decrease in OHU at the equator relative to the extra-tropics, which is responsible for a strengthening of the HC (Feldl & Bordoni, 2016). Due to the meridional forcing structure, which is set by the surface temperature (Jeevanjee et al., 2021) and therefore

enhanced in the tropics relative to the extra-tropics, one finds a positive forcing contribution. The contribution of a feedback is determined by its meridional structure not by it being positive or negative, because it is always evaluated against its meridional mean. A feedback that is larger or smaller than its mean value in the tropics compared to the extra-topics or the poles strengthens or weakens the HC, respectively. For a uniform increase in CO_2 concentration, this is the case for the water vapor feedback, which is amplified in the tropics. The lapse rate, Planck and surface albedo feedback contribute to a weakening of the HC. The cloud feedback shows a large model dependence and can either contribute positively or negatively (Feldl & Bordoni, 2016).

When transferring this to the tropospheric CO_2 case, almost all aspects are expected to be identical here. There only might be some uncertainty again regarding the eddy component. This means the HC response should be similar as for a uniform increase in CO_2, i.e., a weakening of the HC. For an increase in stratospheric CO_2, one would expect no parameter to change significantly, and thus also the HC to stay rather constant. The biggest uncertainty is in the eddy and the forcing component. GMS and the feedback component should be zero because they both only relate to temperature changes in the troposphere, which we do not expect with an increase of stratospheric CO_2.

2.4.1. Other drivers of HC strength changes

In the previous section, three frameworks were introduced that are able provide physical explanations for HC changes. In the literature, several mechanisms more have been proposed to explain HC weakening with greenhouse gas forcing. There seems to be a 'tug of war' between different mechanisms.

There are two mechanisms relating the weakening of the HC to the direct effect of CO_2 increase in the atmosphere. Direct means only considering CO_2 effect on radiative transfer without its influence surface temperature changes with their corresponding feedbacks.

Around 81% of the net radiative cooling of the atmosphere is balanced by latent heating which poses a constraint on the precipitation rate (Hartmann, 2016). Bony et al. (2013) argue that by increasing the CO_2 in the atmosphere, the loss of infrared radiation to space is reduced. This warms and stabilizes the atmosphere weakening upward motions, i.e., the circulation.

The second mechanism employs the spatial structure of CO_2 forcing. In regions of ascending motion, such as the equator, the CO_2 radiative forcing is reduced. This is due to the presence of deep-convective clouds and the high humidity, which both are effective longwave absorbers. Thus, CO_2 below cloud top or the moist atmosphere has a limited effect on OLR. In subsiding regions, such as the subtropics, the CO_2 radiative forcing is larger because this effect is missing. Given a larger forcing in the extra-tropics than at the equator, less energy transport and also a weaker circulation is required. About 15% of the century scale weakening is attributed to the direct effect of CO_2 using fixed-SST simulations (Merlis, 2015).

Another mechanism that explains the HC weakening compares the change in saturation vapor pressure according to the Clausius-Clapeyron equation to the changes in precipitation. The global mean precipitation can be estimated as

$$P = Mq \tag{2.25}$$

where M is the convective mass flux and q is a typical boundary layer mixing ratio, which scales with the CC equation. In its linearized form, one obtains the following expression for

the change in mass flux

$$\frac{\delta M}{M} = \frac{\delta P}{P} - \frac{\delta q}{q}. \tag{2.26}$$

Inserting typical global mean values for $\delta P/P = 0.03\delta T$ and with CC $\delta q/q = 0.07\delta T$ gives a mass flux change of $\delta M/M = -0.04\delta T$, a weakening of global circulations (Held & Soden, 2006). This relationship holds better for global zonal circulations, e.g., the Walker circulation, than the HC (Vecchi & Soden, 2007).

Similiar to the decomposition of HC changes in the energetic framework, diabatic heating, eddy heat flux, eddy momentum flux, zonal friction and static stability changes can also be seen as the drivers of change in maximum MSF (Vallis et al., 2015). None of these factors alone can fully explain the changes but the most important contribution according to this decomposition comes from an increase in static stability with diabatic heating changes even contributing to a strengthening of the HC on longer time scales (Chemke & Polvani, 2021). For a comprehensive evaluation and comparison of the different proposed HC weakening mechanisms, one should consult Chemke and Polvani (2021).

3
Methods

3.1. Model setup

To investigate the HC response under different vertical CO_2 forcings, we used an atmospheric circulation model to simulate their influence on the circulation response. To understand the resulting temperature profiles, we used simulations from the RCE model *konrad* as verification. The HC response is very much dependent on the vertical structure of the CO_2 concentration. We use the different HC frameworks discussed above to diagnose the physical drivers of HC changes. Here, we will describe the specific model setup and the evaluation metrics that we used.

3.1.1. The atmosphere model ECHAM6

For our experiments, we use ECHAM 6 (Stevens et al., 2013), a CMIP 6 type atmospheric General Circulation Model (GCM), coupled to a slab ocean with a depth of $d = 50\,\mathrm{m}$. In the version T63L47, the model has a vertical resolution of 47 pressure levels on a $1.875\,^\circ \times 1.875\,^\circ$ (192×96) grid. We used this complex configuration with continents, orography, water cycle etc. to obtain a realistic circulation response. We prescribed a climatological q-flux that mimics the ocean energy transport of a coupled control run. We fixed the sea ice to avoid problems due to the fixed energy flux in the slab ocean, which would conflict with latent heat release due to the phase changes of the melting ice.

We ran four types of simulations: First, we simulated a pre-industrial control (PI) climate with a CO_2 concentration of 280 ppm. In the second, we abruptly quadrupled the CO_2 concentration throughout the entire atmosphere (*4xCO2 Uniform*). Using the climatology tropopause height (see Section 3.2 for its definition) field, we only abruptly quadrupled CO_2 in the troposphere (*4xCO2 Tropo*) in the third, and in all the layers above the tropopause (*4xCO2 Strato*) while the CO_2 concentration in the remaining domain stays at the PI level in the fourth experiment. The CO_2 concentrations of the four experiments is summarized in Figure 3.1.

To calculate the radiative forcing on the atmosphere, we performed identical simulations as in the slab ocean setup, while fixing the sea surface temperature (SST) to pre-industrial levels. The goal was here to distinguish between the direct effect of an increase in CO_2 in the energetic framework on the HC versus the effect of increasing global temperatures. In our experiments the land is still able to warm but this only accounts for a small amount of the net

forcing. Also fixing land temperatures increases the net forcing by 1 W/m^2 compared to only fixed SSTs and fixed sea ice (Andrews et al., 2021).

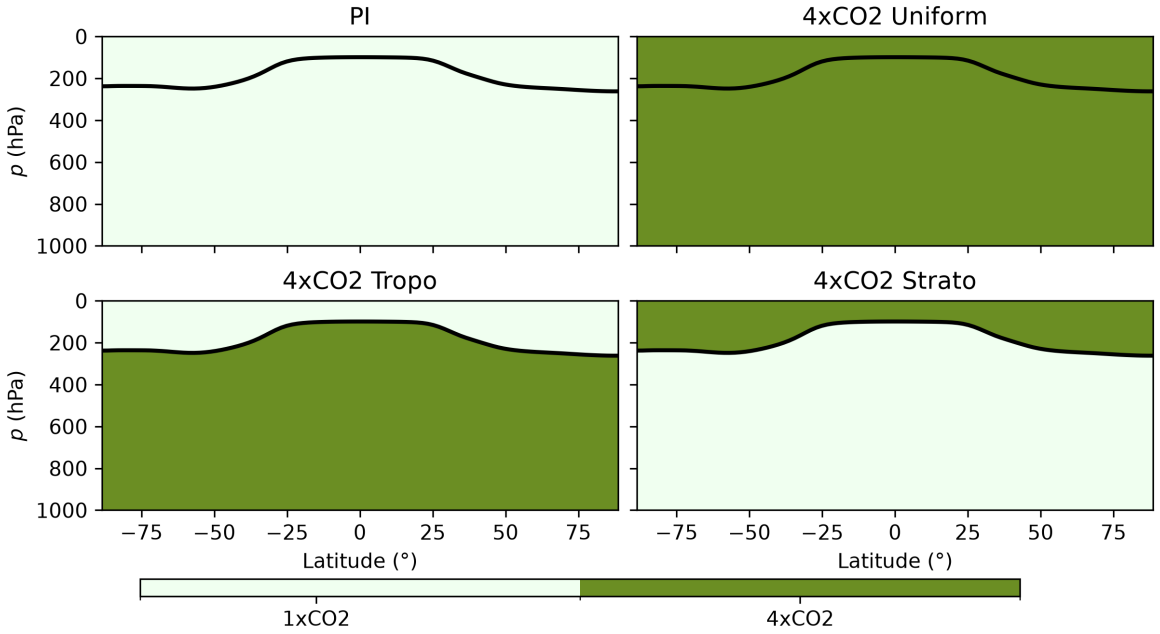


Figure 3.1: The CO_2 concentration in our four experiments. The black line indicates the zonal and temporal mean climatology tropopause height from the pre-industrial control run.

For all analyses we used yearly mean data. The first ten years of every simulation are discarded as spin-up. The data in PI run is then averaged over a period of 30 years, *4xCO2 Uniform* over 40 years, and *4xCO2 Tropo* and *4xCO2 Strato* over 50 years.

3.1.2. The radiative convective equilibrium model konrad

To understand the temperature response in the GCM simulations, we used the RCE model *konrad* (Kluft et al., 2022). It solves the RCE in one dimension, hence it does not account for changes in the atmospheric circulation or horizontal inhomogeneities. As in the experiments with the GCM, we ran four experiments with the one-dimensional vertical separation of CO_2 concentrations as in Figure 3.1. In these simulations, we used the level of the cold point temperature, which is the coldest point in the atmospheric column, as tropopause height. As the surface boundary, we used a slab ocean, which absorbs the TOA radiative imbalance of the PI run in *konrad*. Thus, the surface temperature is not fixed but can adjust to the increase in CO_2 concentration.

3.2. Diagnostics

3.2.1. Hadley cell strength and extent

The HC can be characterized by its strength and extent, both of which can be determined by calculating the meridional mass stream function (MSF)

$$\psi(\phi, p) = \frac{2\pi a \cos\phi}{g} \int_0^p \bar{v} \mathrm{d}p' \tag{3.1}$$

with the radius of the Earth a, the latitude ϕ, the gravitational acceleration g, the pressure p, and the zonal mean meridional wind $\bar{v}$. The MSF is always parallel to the flow in the vertical p–ϕ plane (see Figure 1.1). A positive value of $\psi_{\max}$ indicates a clockwise circulation and a negative value a counter-clockwise circulation. In the annual mean, the northern hemisphere shows a clockwise and the southern hemisphere counter-clockwise circulation. Since the MSF vertically integrates the mass flux, its values symbolize the net magnitude of mass transport above and below and to the right and the left of a certain grid point.

At the HC maximum the integral covers exactly the upper or lower branch of the circulation. We will use the maximum and minimum of annual-mean stream function between 40 °S and 40 °N, and between 200 to 900 hPa for the northern and southern hemisphere HC, respectively. In the literature, one can find several different metrics for the HC strength. Some use the MSF at a fixed pressure level or spatially average, while others use the minimum of the vertical pressure tendencies ω. HC trends have shown a large sensitivity to the metric used (Pikovnik et al., 2022).

We compute the HC extent ϕ_{H} by linearly interpolating the subtropical zero-crossing latitude of ψ averaged between the 300 to 700 hPa levels. All anomalies for a quantity A, such as the MSF ψ or the tropopause height H, if not mentioned differently, are calculated as

$$\frac{\delta A}{A} = \frac{A - A_{\mathrm{PI}}}{A_{\mathrm{PI}}}. \tag{3.2}$$

3.2.2. The axisymmetric scaling parameters

To investigate the drivers of HC changes according to the axisymmetric theory from Section 2.3.1, we linearize Equation (2.4) and Equation (2.5) to

$$\frac{\delta\phi_{\mathrm{H}}}{\phi_{\mathrm{H}}} = \frac{1}{2}\frac{\delta H}{H} + \frac{1}{2}\frac{\delta\Delta_h}{\Delta_h}, \tag{3.3}$$

and

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2}\frac{\delta H}{H} + \frac{5}{2}\frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v} \tag{3.4}$$

assuming that only the tropopause height H, the meridional temperature gradient Δ_h, and the vertical stability Δ_v change in our simulations while all other parameters remain constant. We define these parameters in our model simulation as the following: The tropopause height H is at the lowest level at which the lapse rate decreases to 2 °C km^{-1} or less, provided also that the average lapse rate between this level and all higher levels within 2 km does not exceed

$2\,^{\circ}\mathrm{C\,km^{-1}}$ (World Meteorological Organization, 1992). The result was then averaged over 15 °S to 15 °N.

The definitions of the meridional temperature gradient Δ_h. and vertical temperature gradient Δ_v are adopted from D'Agostino et al. (2017). We use the meridional near-surface potential temperature contrast between the deep tropics (equatorward of 10 °) and the extratropics (40 to 60 °) averaged over 700 to 800 hPa as Δ_h. For Δ_v, we use the potential temperature difference between 700 hPa and 800 hPa averaged between 20 to 40 °. The results were tested also with slightly different definitions, which did not change the results qualitatively.

3.2.3. The integrated eddy momentum flux divergence

For the eddy-driven HC regime, we have to obtain the integrated eddy momentum flux $\mathcal{T}$ from Equation (2.11), we first have to calculate the covariances $\overline{u'v'}$, $\overline{u'\omega'}$ from 6 hourly output data. For this 6 hourly data, we only have an averaging period of 10 years after spin-up. The 6 hourly output is necessary because this is the time scale, on which the eddies occur that influence the mean meridional circulation. Because we use this slightly different dataset also the changes in HC strength and extent are slightly different. To calculate these, it holds that

$$\overline{a'b'} = \overline{ab} - \overline{a}\,\overline{b}$$

with the overline indicating monthly and zonally mean values. To make sure that the deviations are not simply caused by seasonal variations, eddy momentum fluxes are calculated for each month individually and then averaged over all months to get the annual mean data. The integrated eddy momentum flux divergence $\mathcal{T}$ from Equation 2.11 is calculated at every grid point. We compare it to the HC strength at the latitude and pressure level of the maximum MSF $\psi_{\max}$.

3.2.4. Applying the energetic framework

We use Equation 2.24 to calculate the contributions of changes in forcing, feedback, GMS, eddies, and OHU to changes in energy transport. The contribution is always evaluated at the latitude of the MSF maximum in the PI control run even if it has changed in the forced simulations. This is different than how we usually use it (see Equation (3.2)). It guarantees that this decomposition is energetically closed but also leads to slightly different values of HC changes compared to the previous analyses. To calculate the forcing on the atmosphere, we consider the net energy input (NEI) in the fixed SST simulations. The NEI can be determined by evaluating the net TOA fluxes and subtracting the net surface fluxes. It quantifies how much energy has to be transported by the atmosphere due to the direct effect of CO_2.

4

Results

This chapter presents the results of our simulations. The first part covers the temperature and atmospheric circulation response in our experiments, while the second part identifies the mechanism behind the HC responses.

4.1. The temperature and circulation response

4.1.1. Linearity of the decomposition

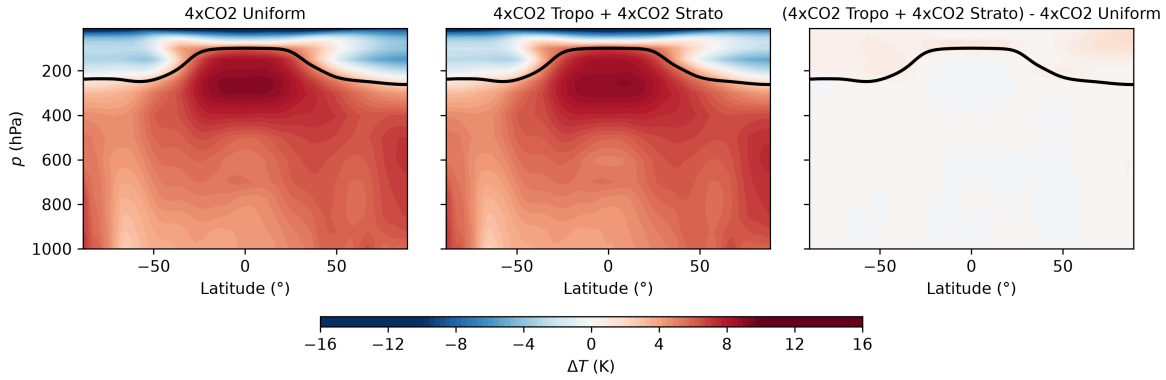


Figure 4.1: The temperature anomalies with a uniform increase of CO_2 concentration, the added *4xCO2 Tropo* and *4xCO2 Strato* cases, and their non-linearities as the difference (from left to right). The black line indicates the tropopause height in the PI run.

To verify that the vertical separation of two different CO_2 concentrations does not show any non-linear behavior, we compare the temporal and zonal mean temperature anomalies of the *4xCO2 Uniform* case to the added up anomalies from *4xCO2 Tropo* and *4xCO2 Strato*. As one can see in Figure 4.1, the non-linear effects in the temperature response are really small. The absolute difference between the sum of the anomalies of *4xCO2 Tropo* and *4xCO2 Strato* minus the anomalies of *4xCO2 Uniform* have a mean value of 0.2 K and a maximum magnitude of 1.8 K. The largest discrepancies can be found in the northern hemisphere stratosphere and right above the tropopause in the tropics.

We find a similar behavior for the circulation response. In Figure 4.2, one can see that the MSF in *4xCO2 Uniform* very much resembles the sum of *4xCO2 Tropo* and *4xCO2 Strato*. Here, the mean absolute difference is about 0.5% with a maximum difference of 6.3% of the maximum absolute value of the MSF in the PI run.

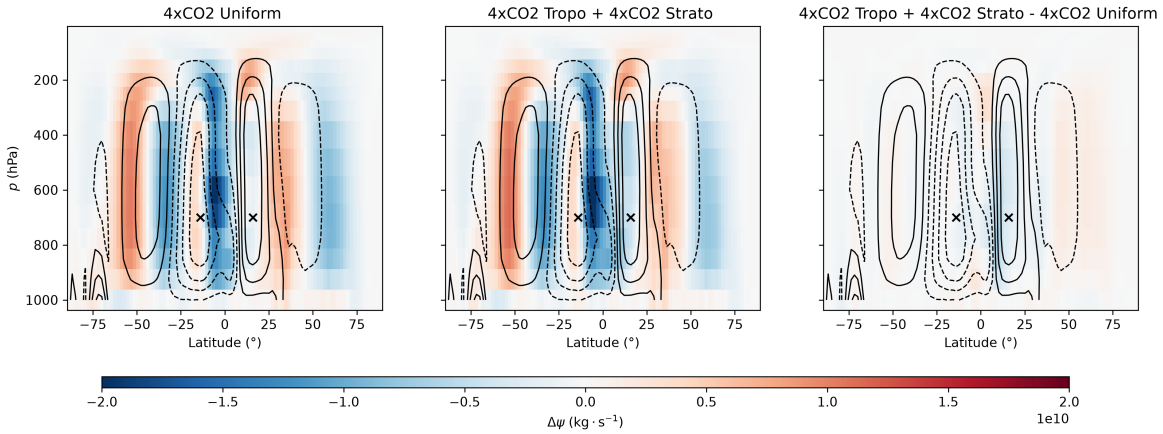


Figure 4.2: Testing the linearity of the circulation response. The shading indicates the anomalies of the MSF in *4xCO2 Uniform*, the added *4xCO2 Tropo* and *4xCO2 Strato* cases, and their non-linearities as the difference (from left to right). The black contours show the MSF in the PI run. Continuous lines are positive (clockwise) while dashed lines indicate negative (counterclockwise) values. The cross marks the location of the MSF maximum in the PI run.

Overall, the decomposition of the CO_2 increase works. The response is approximately linear in the temperature and MSF anomalies. Nonetheless, one should bear the small non-linear effects in mind in the further analysis of these experiments.

4.1.2. Tropospheric warming and stratospheric cooling can be decomposed

Figure 4.1shows that the temperature anomalies in *4xCO2 Uniform* display quite well the expected patterns established in Section 2.1. The temperature throughout the entire troposphere is increased while the warming in the upper tropics is enhanced and amplified in the Arctic. The stratosphere is cooled.

We now expected that the CO_2 increase in *4xCO2 Tropo* is responsible for the warming of the troposphere, whereas the CO_2 increase only in the stratosphere is responsible for stratospheric cooling. This is what we see in Figure 4.3. The warming in the troposphere in *4xCO2 Tropo* maintains the spatial structure from the *4xCO2 Uniform* case. Additionally, in *4xCO2 Tropo* one can notice an additional cooling in the stratosphere right above the tropopause. Besides the cooling of the stratosphere,in *4xCO2 Strato* the tropospheric temperature remains almost constant. Only right above the tropopause, there is an unexpected anomalous warming. This warming/cooling right above the tropopause might have an influence on the tropopause height and the vertical stability, which have been discussed to result in HC changes.

To better understand the cause of the anomalous warming in *4xCO2 Strato* and the anomalous cooling in *4xCO2 Tropo* right above the tropopause, we perform experiments with *konrad* changing the vertical CO_2 concentration. In Figure 4.4, one can see by comparing *4xCO2 Uniform* to *4xCO2 Tropo* + *4xCO2 Strato* that there is also in *konrad* some non-linearity in the response right above the tropopause. However, even when considering the non-linearity

there is an anomalous warming above the tropopause in the *4xCO2 Strato* and an anomalous cooling in the *4xCO2 Tropo* case. In the experiments with *konrad* the warming and cooling at the tropopause also extends below the tropopause. The anomalous heating rates arising at and around the level of discontinuous CO_2 concentrations are induced by the stepwise increase in optical thickness associated with the step wise increase of CO_2 might be its cause. We have tested this hypothesis by linearly increasing CO_2 on some levels below and above the tropopause, which weakens the anomalous heating pattern. From the experiments with *konrad*, we can conclude that the heating in *4xCO2 Strato* and cooling in *4xCO2 Tropo* above the tropopause in the experiments with ECHAM6 are a static thermodynamic response and not evoked by changes in the atmospheric circulation.

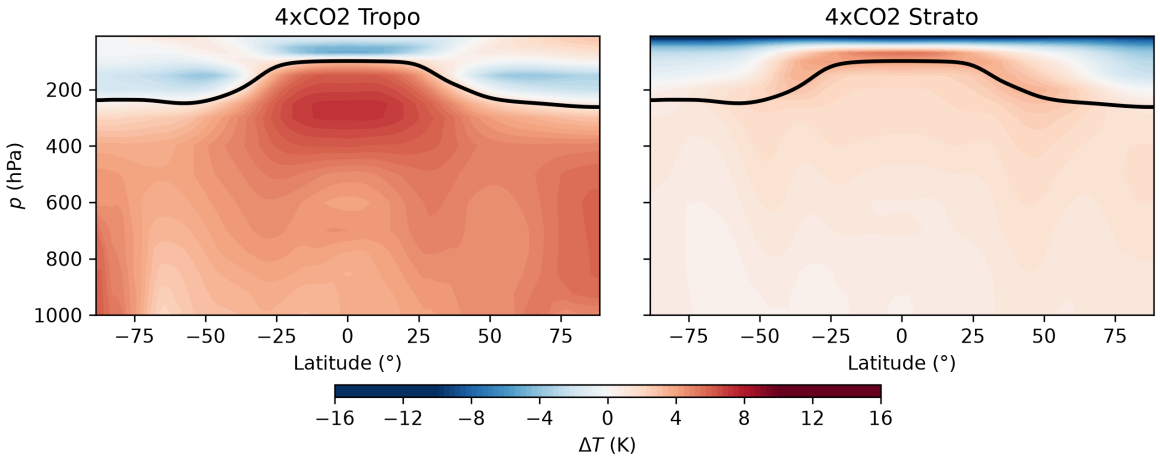


Figure 4.3: The temperature anomalies for the experiments with a GCM where we only increase CO_2 in the troposphere (left hand side) or in the stratosphere and everything above (right hand side). To separate these two cases we use a PI climatology tropopause height (black line).

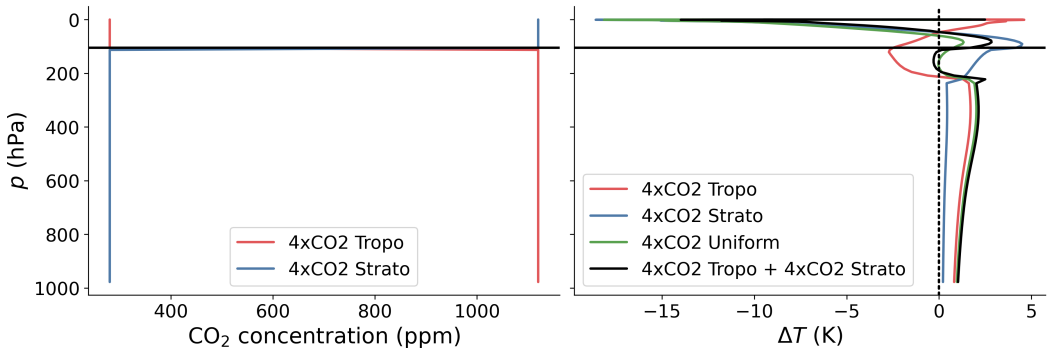


Figure 4.4: CO_2 concentration and temperature anomalies for the experiments with the one-dimensional RCE model *konrad*. The black, horizontal line indicates the tropopause height determined as the cold-point temperature from a PI control run.

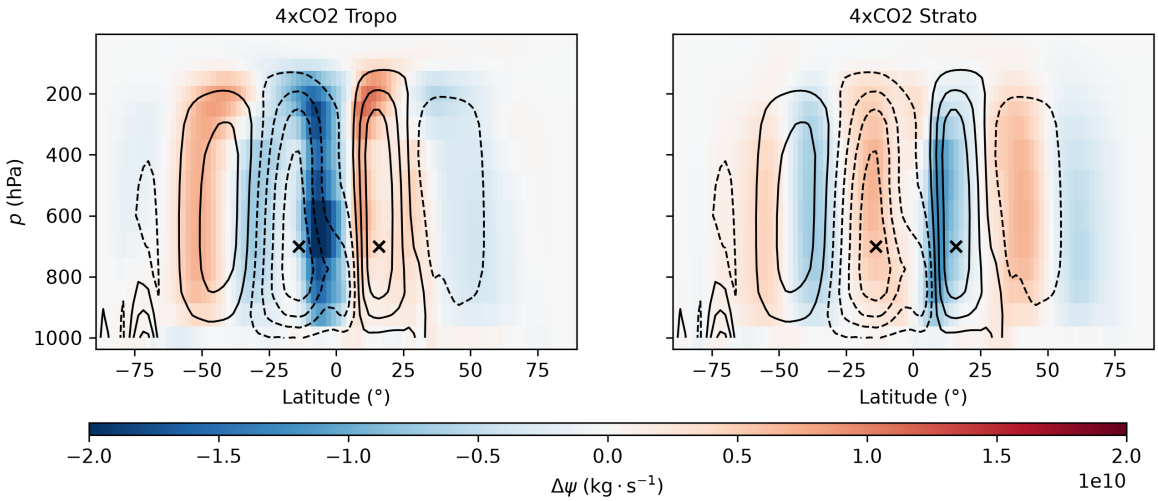


Figure 4.5: The circulation response in the *4xCO2 Tropo* and *4xCO2 Strato* experiment. The shading indicates the anomalies of the MSF and black contours show the MSF in the PI run. Continuous lines are positive (clockwise) while dashed lines indicate negative (counterclockwise) values. The cross marks the location of the MSF maximum in the PI run.

4.1.3. Opposing Hadley circulation responses

Given the temperature responses in *4xCO2 Tropo* and *4xCO2 Strato*, one would still expect the net weakening to originate from tropospheric CO_2 because the associated warming of the troposphere change the temperature gradients and the tropopause height. What we find when analysing the HC extent according to the definitions established in Section 3.2.1 is that the HC is weakening and widening in the *4xCO2 Uniform* case (see Figure 4.2, left panel) in both hemispheres. This comes as a surprise because one would expect an increase in tropospheric CO_2 to be the dominant driver here. Although, also the cooling of the stratosphere is expected to change meridional temperature gradients. Figure 4.5 shows that the annual mean HC is strengthening with *4xCO2 Tropo* and weakening with *4xCO2 Strato*. This suggests that the weakening cannot be traced back to the increase in tropospheric CO_2, but to an increase in stratospheric CO_2. A further implication might be that the cooling of the stratosphere is responsible for the net weakening of the HC in *4xCO2 Uniform* and not the warming of the troposphere. For a summary of all HC changes see Table 4.1.

One should notice here that the HC strength response is not linear in either hemisphere, while the changes in HC extent are to a good approximation. The non-linearity in the northern hemisphere accounts for 3.4% and in the southern hemisphere for 1.5%. Considering the magnitude of the individual changes, which are all significantly larger than the HC response to the uniform forcing and the non-linearities, we perceive them as robust physical responses instead of artifacts due to the non-linearities. The HC strength response in *4xCO2 Strato* is quite different between the hemispheres with the northern hemisphere weakening almost twice as much as the southern. In the next section, we identify the underlying physical mechanisms that lead to a strengthening in *4xCO2 Tropo* and a weakening in *4xCO2 Strato* employing different HC frameworks.

	4xCO2 Uniform	*4xCO2 Tropo*	*4xCO2 Strato*
$\delta\psi_{\text{max,NH}}$	−1.9%	3.6%	−8.9%
$\delta\psi_{\text{max,SH}}$	−2.5%	4.1%	−5.1%
$\delta\phi_{\text{H,NH}}$	1.6 °	0.7 °	0.7 °
$\delta\phi_{\text{H,SH}}$	−1.6 °	−1.0 °	−0.6 °

Table 4.1: The changes in HC extent and strength in the experiments for the northern and southern hemisphere. Negative values for the southern hemisphere extent indicate a widening of the HC.

4.2. Physical drivers of HC strength response

4.2.1. Predictions from axisymmetric scalings

Trying to explain the opposing Hadley cell strength responses but also the HC widening in both cases we use the axisymmetric scalings from Section 2.3.1. The following analysis is very much similar to the one from Seo et al. (2014). Given the temperature response described above, it is not possible to infer a qualitative explanation of the HC changes from the temperature field using the axisymmetric scalings (see also Section 2.3.1). The axisymmetric scalings predict that the HC strength and extent are a function of the tropopause height H, the meridional temperature gradient Δ_h, and the vertical stability Δ_v. These parameters are defined according to Section 3.2. The individual predictive power of each of these parameters is displayed in Figure 4.6. For the HC extent anomalies, we find the best correlation ($R^2 = 0.89$) with the change in vertical stability, a parameter which according to Equation 2.4 is not supposed to influence the extent of the HC at all. We find a smaller correlation ($R^2 = 0.67$) between δ_h and the HC extent with the tropopause height playing a negligible role ($R^2 = 0.24$). For the correlations of the parameters with the HC strength we find even less agreement. The largest correlation exists for the tropopause height ($R^2 = 0.61$). The meridional and vertical temperature gradients show almost no skill at all. The increase in tropopause height in *4xCO2 Tropo* has the implication that a small part of the "new" troposphere relative to the PI simulation is not affected by the quadrupling of CO_2. The same is true for the decrease of tropopause height in the *4xCO2 Strato* case. Since the individual parameters cannot significantly predict the HC changes, we now consider their combined contribution. Using the linearisations from Section 3.2.2, we can calculate what the change in HC would be by just considering the change of the tropopause height, the meridional temperature gradient, and the vertical stability. In Figure 4.7, one can see the combined contributions according to the axisymmetric scalings. If the scalings hold exactly, all datapoints would lie on the one-to-one line. We see that the changes in the thermodynamical parameters significantly underpredict the simulated changes in HC extent while they overpredict the changes in HC strength. Thus, one can draw the conclusion that the axisymmetric scalings are not suitable to explain the changes in the HC that we see in the simulations.

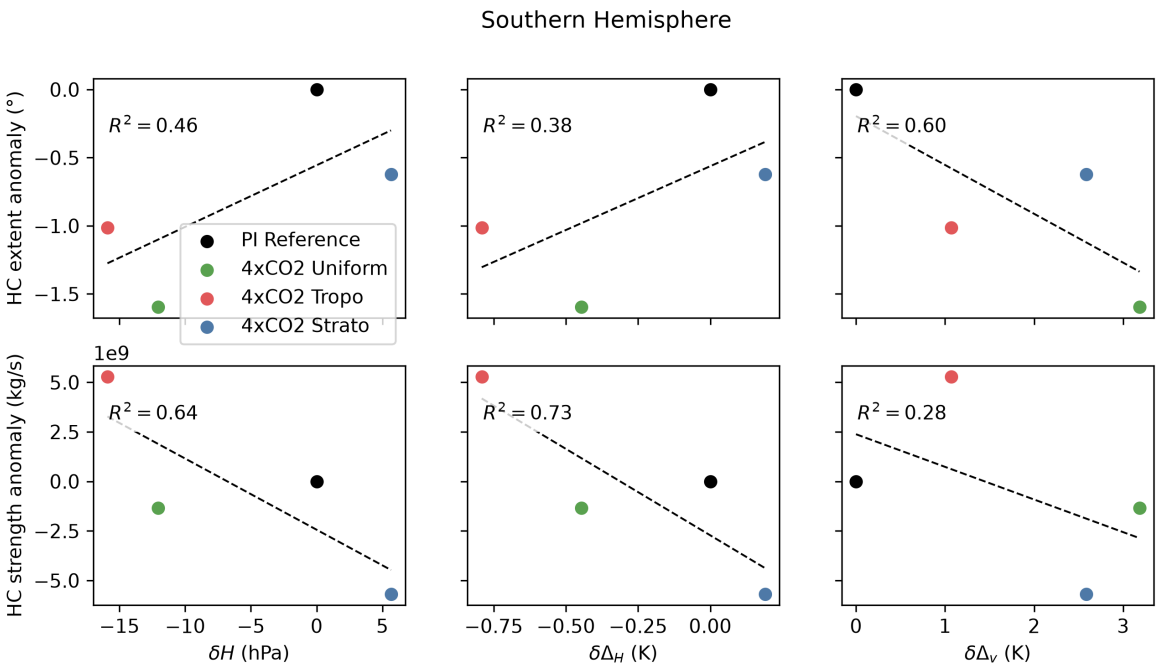


Figure 4.6: Changes in annual mean HC extent and strength against changes in tropopause height H, meridional temperature gradient δ_h, and vertical stability δ_v (top down) for the four experiments. Linear regression shows their agreement.

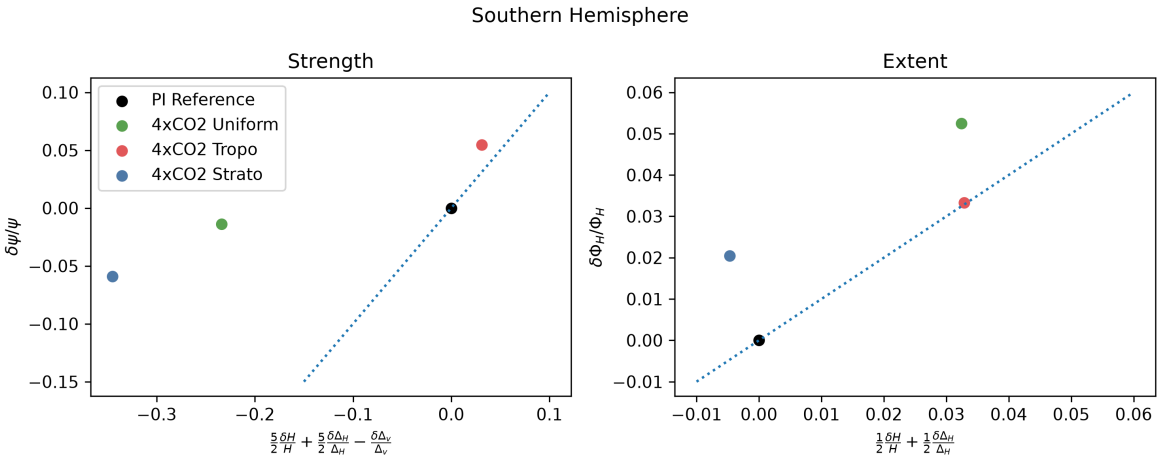


Figure 4.7: HC changes against their prediction from the linearised axisymmetric scalings for the four experiments. A good agreement is given for datapoints on the one-to-one line.

Doing the same analysis for the southern hemisphere (not shown), we find different R^2 values but not an overall increase in predictive skill of the axisymmetric scalings. Also when investigating the winter HCs, which, according to the dynamical framework, are closer to angular momentum conservation, the results do not change qualitatively. This has the following reasons too few data points to do an actual fit, but also in the paper they do not find a convincing relationship Finally, we suspect in particular the influence of the eddies to

fundamentally influence the strength of the HC. Thus, we investigate the changes in eddy momentum flux in the next section.

4.2.2. Eddy driven HC changes

As was reported in Section 2.3.2, eddies are known to substantially influence the mean meridional circulation. This has been shown in particular for the HC in its equinox, which is similar to the annual mean HC, and summer regime. The flow in our simulations has a maximum Rossby number of about 0.3, which favors this perspective over the axisymmetric one. Since the eddy fluxes are closely related to the baroclinicity in a flow, which itself is driven by meridional temperature gradients, we expect them to change given the very different temperature response in *4xCO2 Tropo* and *4xCO2 Strato*. An increase in the meridional temperature gradient strengthens the zonal wind and thus the generation of eddies. In particular, in *4xCO2 Tropo* the meridional temperature gradient at the height of the subtropical jet stream increases while in *4xCO2 Strato* it decreases (see Figure 4.3). Using Equation (2.11) and calculating the eddy momentum fluxes as described in Section 3.2.3, we can estimate to what extent the HC in the simulations is driven by eddy momentum fluxes and their relative contribution to HC changes.

Plotting the maximum MSF (HC strength) against the integrated eddy momentum flux $\mathcal{T}$ as in Figure 4.8 reveals that the annual mean northern hemisphere HC is generally more eddy-driven than its southern hemisphere counterpart. If the HC would be entirely eddy-driven all datapoints would lie on the one-to-one line. This can be seen for the Ferrel cell (not shown). Changes in eddy momentum flux are similar in both hemispheres. Relative to the PI simulation $\mathcal{T}$ is increasing in *4xCO2 Tropo*, slightly weakening in *4xCO2 Uniform* and largely weakening in *4xCO2 Strato*. The relative changes in $\mathcal{T}$ overpredict the changes in HC strength in the northern hemisphere *4xCO2 Tropo* and *4xCO2 Strato* as well as in the southern hemisphere *4xCO2 Uniform* and *4xCO2 Strato* cases. This indicates that more factors that drive the HC and influence their relative changes should be considered on the analysis. Again, one can notice the non-linearity of the response, with the sum of the changes in $\mathcal{T}$ in *4xCO2 Tropo* and *4xCO2 Strato* exceeding the *4xCO2 Uniform* case. However, performing a linear regression shows a good correlation between the integrated eddy momentum fluxes and the HC strength.

TODO indirect thermal drivers

4.3. Energetic perspective on the HC strength response

The analysis using angular momentum budget presented in the previous sections is not able to explain, why tropospheric CO_2 strengthens and stratospheric CO_2 weakens the HC. The simulated annual mean HC is neither in the mean angular momentum conserving limit nor in the eddy-driven limit. For intermediate regimes, there does not exist a theoretical framework. The energy budget framework provides a diagnostic perspective on HC strength changes. In the following, we will decompose the HC strength changes into contributions from changes in GMS, energy eddy fluxes, OHU, radiative forcing, and climate feedbacks. We demonstrate the method for the southern hemisphere, where the artificial ocean heat uptake caused by the fixed sea ice is smaller (see Section 4.3.3), and discuss the northern hemisphere afterwards. Southern hemisphere HC changes are slightly changed ($\sim 0.5\%$ in

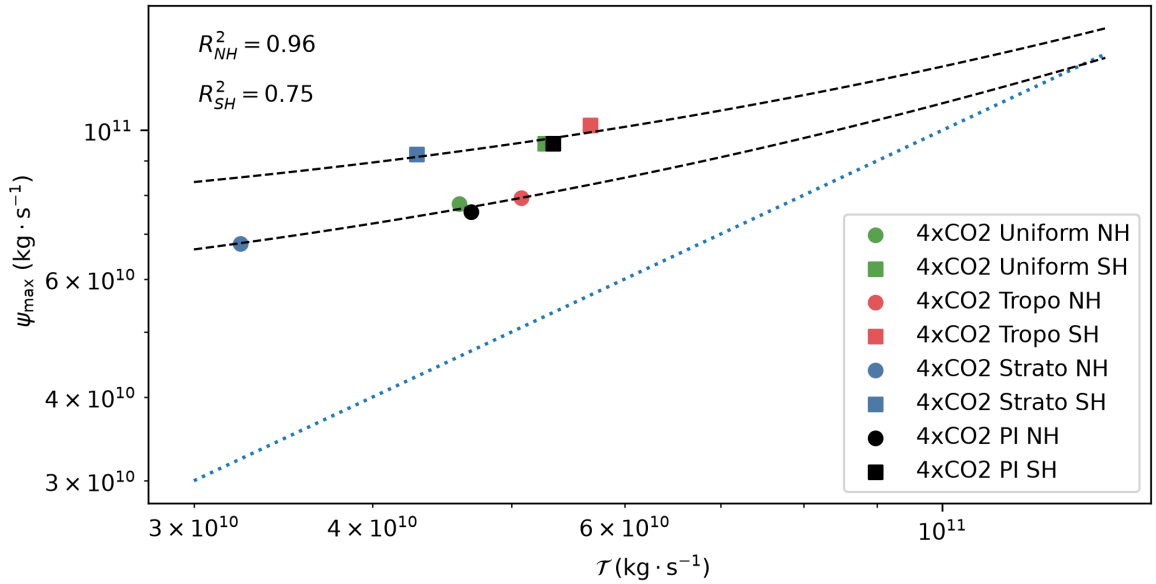


Figure 4.8: The annual mean HC strength as a function of the integrated eddy momentum flux. For an entirely eddy-driven HC, all datapoints are located on the (blue) one-to-one line. For these simulations, this is not the case. However, there is a clear correlation for each hemisphere.

4xCO2 Tropo) in the following analysis compared to previous sections because the latitude of the maximum has slightly moved. We stay at the same latitude so that the calculations stay energetically closed (TODO).

4.3.1. Gross moist stability contribution

The strength of the HC, which is defined by the maximum mass stream flux, is directly related to the energy transport by the HC and its efficiency ($\psi_{\max} = F_{\mathrm{HC}}/\mathcal{H}$). The efficiency of energy transport is described by GMS (see Section 2.4). Increasing the required energy transport by the atmosphere, while the GMS remains constant, also increases the HC strength. If the energy transport by the HC remains constant, but the efficiency of energy transport by the HC $\mathcal{H}$ increases, the HC strength weakens. In this section, we calculate the contribution of changes in HC energy transport F_{HC} and GMS $\mathcal{H}$ according to Equation (2.15) and (2.18) using the linearised energy budget equation.

The bars in Figure **??** display their contributions to changes in HC strength. This bar plot and all following have to be interpreted as follows: The contributions to the right of the dashed vertical line are supposed to add up to the changes in $\psi_{\max}$. The HC changes that cannot be explained by changes of either of these parameters are listed as the residual. The sum of *4xCO2 Tropo* and *4xCO2 Strato* for each contributor is indicated as horizontal line in the *4xCO2 Uniform* bar.

In *4xCO2 Uniform* and *4xCO2 Tropo* the increase in required energy transport is contributing to a strengthening of the HC. The increase in GMS is contributing to a weakening of the HC. For *4xCO2 Uniform* the GMS increase contribution prevails, leading to a small net weakening of the HC. In *4xCO2 Tropo* the changes in F_{HC} are large, leading to the net strengthening of the HC. In *4xCO2 Strato* we see the opposite response. The energy transport and the GMS are

both decreasing. Since the decrease in GMS is smaller than the decrease in energy transport the HC weakens.

To find out what drives the changes in GMS, we plot the vertical profile of changes in MSE flux weighted by the HC strength at the location of the MSF minimum (see Figure 4.9). Vertically integrating these profiles and dividing by the GMS of the PI control yields the changes in GMS. The dominant change in MSE flux occurs at the tropopause, leading to the large net increase in *4xCO2 Uniform* and *4xCO2 Tropo*. This vertical distribution also indicates that the heating in *4xCO2 Strato* and cooling in *4xCO2 Tropo* in the stratosphere, which were discussed in Section 4.1.2, do not affect the changes in GMS. The changes of MSE in the stratosphere seem to leave the GMS unaffected. Thus, integrating the MSE flux, which could be proposed as a solution to circumvent this problem, to the tropopause only does not qualitatively change the result. We further decompose the changes in energy transport, to investigate its drivers in the next sections.

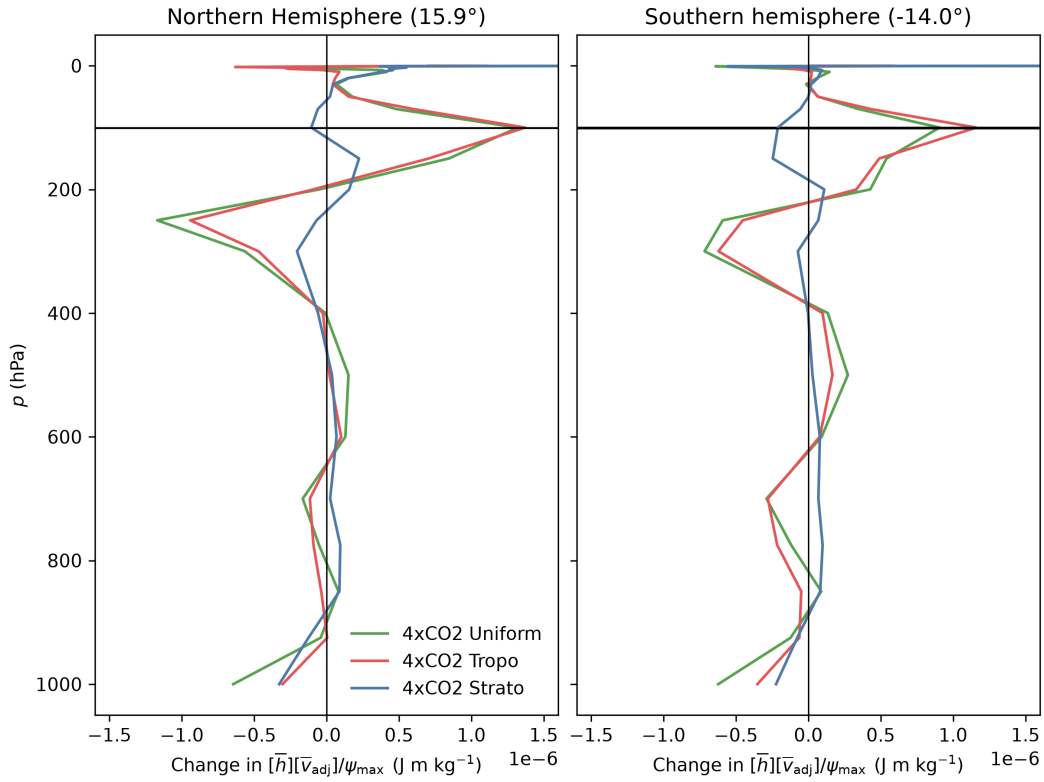


Figure 4.9: Vertical profiles of changes in the meridional MSE flux per unit mass flux for *4xCO2 Uniform*, *4xCO2 Tropo*, and *4xCO2 Strato* at the location of the southern hemisphere MSF minimum and the northern hemisphere MSF maximum. The horizontal black line indicates the tropopause height at this latitude.

4.3.2. Eddy energy flux contribution

The energy transport of the HC can change either by changes in total atmospheric energy transport or by changes in energy eddy transport ($F_{HC} = F_a - F_e$). The eddy term consists of the stationary and transient eddies, because the eddy component is calculated as the residual

of the energy transport of the atmosphere minus the transport by the Hadley cell. The more energy transport by the eddies, the less has to be performed by the mean circulation given the same amount of total energy transport by the atmosphere.

Figure **??** shows that the energy transport by the eddy energy fluxes increases contributing to a weakening of the HC in all experiments. The increase is largest in *4xCO2 Uniform* (17.5%), slightly smaller (13.8%) in *4xCO2 Tropo* and very small (2.6%) in *4xCO2 Strato*. Connecting the increase in eddy energy transport to changes in meridional temperature structure proves difficult. For *4xCO2 Uniform* and *4xCO2 Tropo*, the changes in zonal mean zonal wind in Figure 4.10 clearly show an increase in the subtropical jet strength that coincides with a rise in meridional temperature gradient. This increases the baroclinicity ,i.e., the generation of eddies. For *4xCO2 Strato*, there is a decrease of the mean zonal wind right at the tropopause, but it is increasing in the surrounding areas. The changes of energy transport in the atmosphere show a large contribution to HC strengthening of about 30% in *4xCO2 Uniform* and *4xCO2 Tropo*, while they contribute to a modest weakening of the HC in *4xCO2 Strato*. They will be further decomposed in the next section.

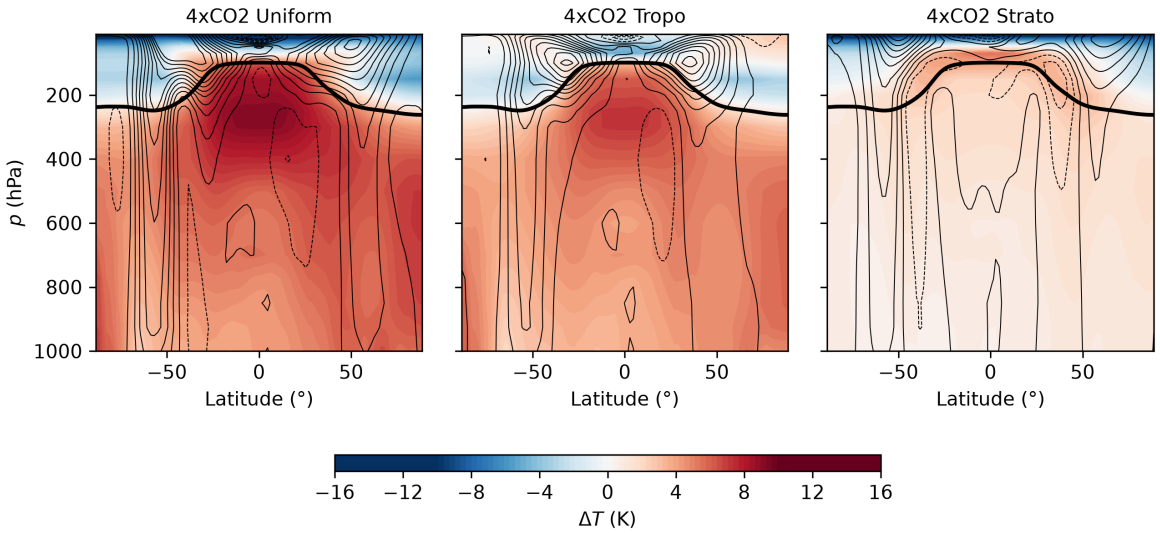


Figure 4.10: The zonal mean temperature anomalies (shading) and the contours of zonal mean zonal wind anomalies at 1 m s^{-1} distance. The thick black line indicates the zonal mean climatology tropopause height from the PI run.

4.3.3. Ocean heat uptake contribution

In our experiments, we are using an slab-ocean configuration with a prescribed energy flux at the surface. Thus, one does not expect any changes on ocean heat transport. When investigating the surface fluxes (net LW, net SW, latent heat, sensible heat, see Figure 4.11), we find that this is not true for the simulated time period.

The culprit is the sea ice. Looking at Figure 4.11,one can see that the anomalies in surface heat fluxes have an identical spatial structure as the sea ice extent. Since the sea ice stays fixed in our simulations the SSTs below can only warm through conduction. With the air temperature above the sea ice remain much larger than the SSTs below the sea ice. There

is a constant flux of energy out of the atmosphere into the ocean. This affects the global atmospheric circulation, because the removal of energy in higher latitudes requires additional poleward energy transport by the atmosphere. Running the simulations for a longer period until the temperatures above and below sea ice are in equilibrium should remove this effect.

Calculating the difference between the energy transport by the atmosphere and the total energy transport from TOA radiative fluxes, we obtain the OHU contribution. Additional poleward transport as given due to the anomalous surface fluxes, requires a strengthening of the HC, which is why the OHU contributes positively in *4xCO2 Uniform* and *4xCO2 Tropo* in Figure **??**. Because the troposphere is not warming, this effect is negligible in *4xCO2 Strato*. Thus, there is no temperature contrast between the surface air temperature above and the SST below the sea ice. The OHU effect is much stronger in the northern hemisphere because there is more sea ice, and more warming of the air close to the surface in the Arctic than in the Antarctic. The contribution of this effect, is of similar magnitude as the strengthening of the HC in *4xCO2 Tropo*. Thus, it might be a large contributor but cannot solely explain the not weakening of the HC in *4xCO2 Tropo*. The OHU at the poles also dampens the weakening response in *4xCO2 Uniform*, which would show a stronger weakening otherwise.

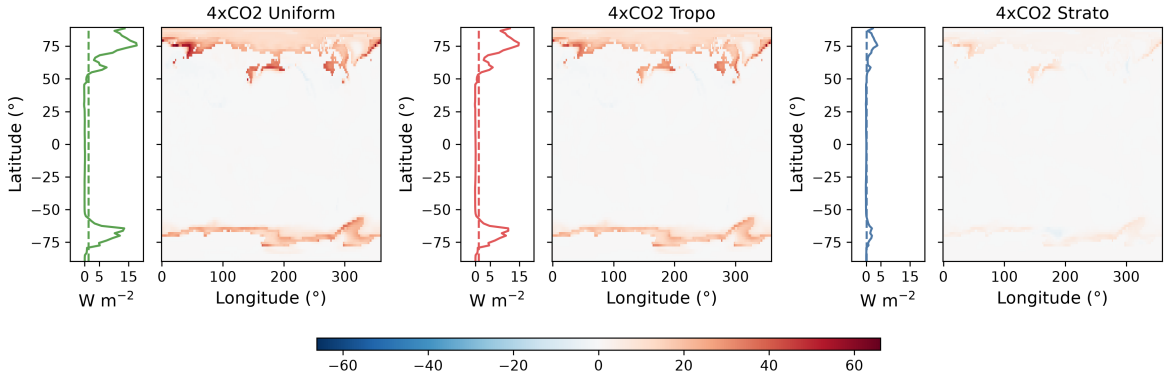


Figure 4.11: Annual mean surface flux anomalies for *4xCO2 Uniform*, *4xCO2 Tropo*, and *4xCO2 Strato*. The map on the right hand side displays the spatial distribution of OHU while the profile plots show the zonal mean for each experiment.

4.3.4. Radiative forcing and feedback contributions

The changes in total energy transport F_{ao} are calculated from the TOA radiative imbalance R_{TOA}. The radiative imbalance can be split up into radiative flux changes due to changes in surface temperature (feedback component) and due to the direct effect of CO_2 (forcing). As forcing we will consider the net energy input (NEI) into atmosphere in the fixed SST simulations. It can be calculated as the difference between fluxes TOA and at the surface. Other studies only use the TOA radiative imbalance here. The meridional structure of the forcing and the feedbacks can also contribute to HC changes. A larger forcing in the tropics compared to the extratropics requires additional poleward energy transport.

This is the case in *4xCO2 Uniform* and *4xCO2 Strato* in the southern hemisphere. Figure 4.13 shows that the spatial structure of the forcing is primarily driven by the distribution of land masses: Since the land cannot take up heat but the oceans can, the areas over the ocean take up more heat than enters the atmospheric column ($R_{surf} > R_{TOA}$), while the energy

that areas over land receive is also partly taken up by the ocean ($R_{surf} < R_{TOA}$). Fixing also the land temperatures increases this effect, because there would be even less OLR at TOA increasing the imbalance. In *4xCO2 Uniform* and *4xCO2 Tropo* the continental land masses of southern Africa, Oceania, and South America are responsible for the zonal mean net surplus of energy relative to the southern hemisphere mid-latitudes. In *4xCO2 Strato* this effect is particularly large in the Antarctic, which is why the forcing contributes to a weakening of the HC here.

The full decomposition of HC strength drivers for the southern hemisphere can be seen in Figure 4.12. Changes in due to feedback only contribute to circulation changes in *4xCO2 Uniform* and *4xCO2 Tropo*. They contribute to a strengthening of the HC, which might have its cause in the tropic amplified meridional structure of the water vapor, and possibly cloud feedbacks. In *4xCO2 Strato* there are almost no changes in surface temperature which is reflected in its low contribution. A further decomposition in the individual feedbacks requires the radiative kernel technique (Soden et al., 2008).

In the southern hemisphere of *4xCO2 Uniform*, the forcing, feedback, and OHU contribute to a strengthening of the HC with the feedbacks having the largest weight. Still, there is a net reduction in HC strength, which is caused by the increase in GMS and eddy energy transport. For *4xCO2 Tropo*, we find very much similar contributions as in the *4xCO2 Uniform* case. The largest differences between these two can be found in the radiative forcing and eddy contributions, which adds up to the net strengthening of the HC in the *4xCO2 Tropo* case. The signal in *4xCO2 Strato* is generally much smaller for each of the contributor. Most of this can be explained by the absence of a temperature response in the troposphere. The largest contribution to the net weakening originates from the radiative forcing contribution. The residual remains small for all cases.

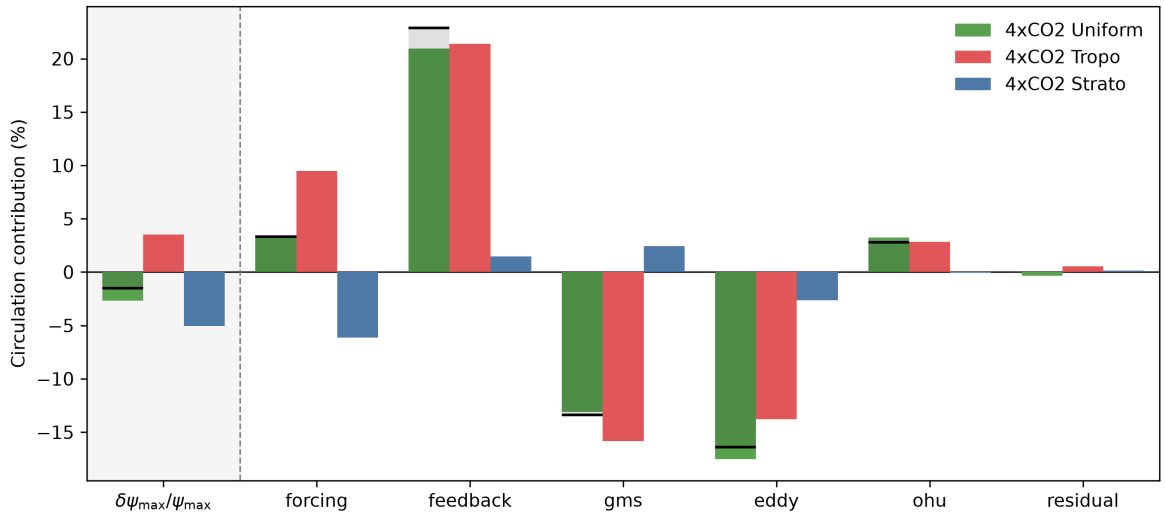


Figure 4.12: Decomposition of changes in total energy transport by the atmosphere F_{ao} to changes due to radiative forcing and changes due to feedbacks for *4xCO2 Uniform*, *4xCO2 Tropo*, and *4xCO2 Strato* for the southern hemisphere.

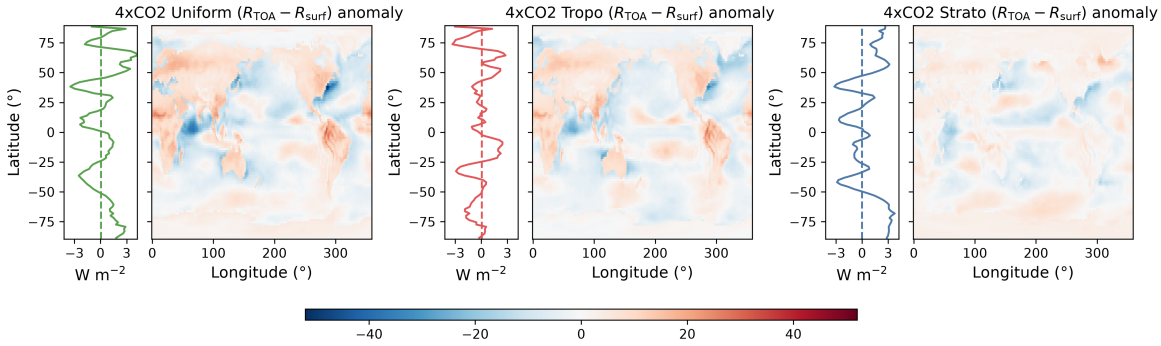


Figure 4.13: Annual mean surface flux anomalies for *4xCO2 Uniform*, *4xCO2 Tropo*, and *4xCO2 Strato* with fixed SSTs. The map on the right hand side displays the spatial distribution of OHU while the profile plots show the zonal mean for each experiment.

4.3.5. Quantifying HC change contributions for the northern hemisphere

When comparing the previously discussed contributions to the HC changes in the southern hemisphere to the northern hemisphere (see Figure 4.14), one can notice major differences. While the forcing contributed positively to the HC strength in *4xCO2 Uniform* and *4xCO2 Tropo* in the southern hemisphere, it contributes to a weakening in the northern hemisphere. Figure 4.13 shows that the large increase of net energy input into in mid-latitudes is produced by the large Eurasian and North American land mass. The contributions of the feedbacks are, as in the southern hemisphere. positive. Only in the *4xCO2 Strato* case, it causes a small weakening. For the contribution of GMS, *4xCO2 Tropo* is much less positive while *4xCO2 Strato* is much more positive. Here, we can also see the largest non-linearity of the sum of *4xCO2 Tropo* and *4xCO2 Strato* which might als cause the large net linearity in the HC changes. For the eddies, we notice that the increase in *4xCO2 Tropo* exceeds the one in *4xCO2 Uniform*, which was not the case in the southern hemisphere. The eddy contribution of *4xCO2 Strato* is even positive, which can be explained by the stronger weakening of the zonal mean zonal winds in the northern hemisphere (see Figure 4.10). The increased OHU has a much larger HC strengthening contribution in the northern hemisphere. As one can see in Figure 4.11, this can be explained by the the larger sea ice fraction and the larger warming in the Arctic, which strengthens the temperature difference between the air and the sea. The contribution to a strengthening of the HC in *4xCO2 Tropo* even outweigh its net strengthening. This means, that without the OHU in our simulations, we would not find a strengthening of the HC in the northern hemisphere. However, the OHU component is even larger in *4xCO2 Uniform*, which show a weakening of the HC strength. In summary, the net response in HC strength is qualitatively similar in the northern and southern hemisphere, the underlying physical mechanism of the response differ significantly.

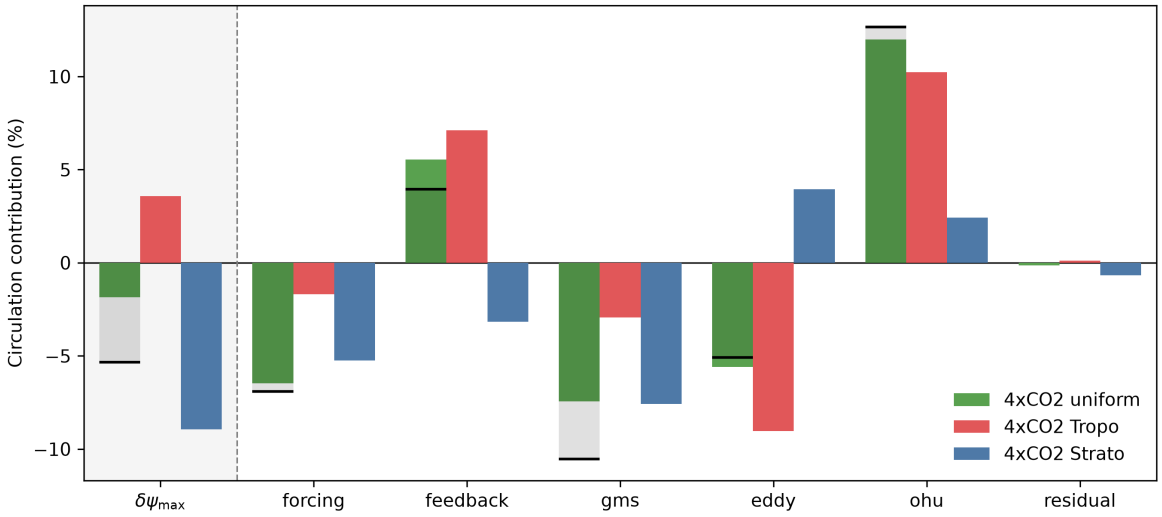


Figure 4.14: Contributions of changes in energy transport due to forcing, feedbacks, eddies,OHU, and GMS to changes in northern hemisphere annual mean HC strength in *4xCO2 Uniform*, *4xCO2 Tropo*, and *4xCO2 Strato*.

5

Discussion and conclusion

5.1. Summary of the main findings

The first part addressed how the atmosphere responds to an increase in tropospheric CO_2 compared to an increase in stratospheric CO_2. We found the most striking differences in the HC response. In our experiments the annual mean HC is strengthening with a quadrupling of CO_2 only in the troposphere, while it is weakening with a quadrupling of CO_2 in the stratosphere. This holds in both hemispheres. For a uniform 4xCO_2 experiment we find a weakening of the HC. This suggests that the CO_2 increase in the stratosphere is responsible for the net weakening of the HC. In all experiments the HC is widening.

The temperature response agrees broadly with our expectations. Tropospheric CO_2 increase causes tropospheric warming, whereas the stratospheric CO_2 increase causes stratospheric cooling. Additionally, we find a modest warming and cooling right above the tropopause in *4xCO2 Strato* and *4xCO2 Tropo*, respectively. Simulations with a RCE model suggest that this is caused by the step-like increase in CO_2 concentration at the tropopause and the thereby induced heating rates.

This raises concerns about the linearity of the circulation and temperature response. We find that the annual mean temperature and circulation response is approximately linear. However, we find a non-linearity in the HC strength (MSF maximum) of 3.4% in the northern hemisphere and 1.5% in the southern hemisphere.

The second part of the analysis discusses the underlying physical mechanisms of the opposing HC strength responses. We find that the scaling laws for the HC strength and extent obtained from the axisymmetric theory cannot explain the HC changes in our simulations. The scaling factors, the tropopause height H, the meridional temperature gradient Δ_h, and the vertical temperature gradient Δ_v, do not demonstrate a good correlation with the simulated HC changes individually. Also combined via the linearised scaling laws they lack predictive power. The validity of the axisymmetric theory is confined to a very idealized setup, which is not given in our simulations. In particular, it neglects the eddies, which are important for controlling the HC strength.

However, also assuming an eddy-driven HC cannot fully explain the simulated changes. The changes in integrated momentum flux show good agreement with the changes in HC strength. A weakening in HC strength coincides with a weakening in integrated eddy momentum flux (high correlation). Still, the angular momentum eddies fail to fully explain the magnitude of the HC strength changes.

Finally, the energy budget decomposition can provide a quantitative explanation for the opposing HC strength changes. HC changes are split up in contributions from the meridional structure of the radiative forcing, climate feedbacks, and ocean heat uptake as well as changes in gross moist stability and eddy energy fluxes.

Although the HC strength response is qualitatively similar, we find large quantitative differences between the hemispheres. In the southern hemisphere, we find the main driver for the strengthening of the HC in *4xCO2 Tropo* to be the meridional structure of the feedback. The meridional structure of the forcing on the atmosphere, is responsible for the weakening in *4xCO2 Strato* in the southern hemisphere. In contrast, for the northern hemispheric strengthening of the HC in *4xCO2 Tropo* the OHU in the Arctic is most important. The increase in GMS is the strongest driver of the HC weakening in *4xCO2 Strato*. However, none of these drivers alone can explain the HC response. Rather, changes in HC strength are the result of a combination of forcing, feedback, GMS, eddy, and OHU contributions.

5.2. Limitations

The large differences between the drivers of HC changes in the northern and southern hemisphere question the generalisability of the analysis. The simulation results may specifically depend on the underlying model setup. Fixing the sea ice fraction has very large implications on the northern hemisphere energy budget. It imposes an energy flux out of the atmosphere in the regions that are covered by sea ice. In the northern hemisphere, this effect can be made responsible for the strengthening of the HC in *4xCO2 Tropo*. In the southern hemisphere, the strengthening barely survives. In both hemisphere, the weakening in *4xCO2 Uniform* would be turned into a strengthening.

Furthermore, the artificial separation of CO_2 concentrations causes non-linear responses in the temperature and wind fields. One also has to consider the non-stationary of tropopause height. Throughout the seasonal cycle the tropopause height changes, which in our case has the consequence that the CO_2 concentration is also increased at locations where we were not initially intended. On top of that, the tropopause height climatology also changes in the forced experiments. For example, in *4xCO2 Tropo* the tropopause height increases relative to the PI control climatology. This has the consequence, that not the entire troposphere is actually exposed to a quadrupling of CO_2.

In our analysis of the drivers of the HC strength response, we have to acknowledge the weaknesses of axisymmetric theory. In the derivation of the HC scalings, it is assumed that our domain is axisymmetric in the zonal direction (no eddies), there is no hydrological cycle, no land masses, no cloud parameterisation, etc. - in short, it is entirely missing the complexity of our model setup. In addition, when we analyse the correlation between the HC changes and the scaling parameter, we only use the four datapoints of their annual mean values, which is a pretty poor data basis. The same limitations are also true for the analysis of angular momentum eddies as driver of HC strength changes.

A limitation of the energetic framework is that we can only use it for diagnostics without explaining why there is a Hadley circulation in the first place. It also fails to provide us with a mechanistic understanding of the processes in the atmosphere but rather stays descriptive in its explanations. Our decomposition tells us which terms balance the change in MSF but not which processes, e.g., change in cloud fraction or the position of the storm tracks, cause

it. There is no direct causal relationship. Changes in the temperature structure affect the GMS which affects the atmospheric circulation but also the circulation affects the temperature structure and thus the GMS back. Eventually, it lacks an explanation for the HC extent drivers.

The experiments have only been performed with a single GCM. The circulation response might show a model dependency. In particular, the cloud feedback shows a large intermodel spread, which has an influence on the net feedback contribution (Feldl & Bordoni, 2016). Also the magnitude of HC strength changes itself, as well as the magnitude of tropospheric warming and stratospheric cooling is different across models.

5.3. Comparison with previous studies

First things first, there is, to our knowledge, no previous study that investigates the impact of a vertical separation of CO_2 increase. Previous studies have only performed a separation of CO_2 concentration in the horizontal (Kim et al., 2022).

In this study we manage to reproduce the weakening of the HC under uniform CO_2 forcing that has been found in model intercomparison studies (see Section 2.2). Also the structure of temperature anomalies is consistent with the literature.

Using the axisymmetring scalings from Section 2.3.1, Seo et al. (2014) finds that the HC strength is significantly correlated with the meridional gradient in potential temperature ($R = 0.57$). For the vertical temperature gradient and the tropopause height they find almost no correlation. D'Agostino et al. (2017) find a consistent relation between the HC strength and the subtropical static stability, and the tropopause height. Using our simulations, we can find neither of these correlations to show a sufficient correlation.

Fan and Dommenget (2024) find the change in tropopause height to have the largest contribution to changes in HC strength. In our simulations we even find opposing responses. The tropopause height in *4xCO2 Uniform* is increasing and the HC strength decreasing. In *4xCO2 Strato* however, the HC strength is also decreasing with the tropopause height is decreasing.

Walker and Schneider (2006) find that the HC strength is approximately equal to the integrated eddy momentum flux in an idealized GCM. One would not expect that this can be easily reproduced in a more realistic GCM. However, we do not find an equality between the integrated eddy momentum flux and the HC strength but a robust linear linear relationship.

Comparing the energy budget decomposition in ECHAM6 in the configuration described in Section 3.1.1 with the MPI-ESM-LR run in Feldl and Bordoni (2016)[1], we find a qualitative agreement. All signs agree with the southern hemisphere energy budget decomposition in *4xCO2 Uniform*, while the individual contributions differ largely.

5.4. Relevance beyond the experiment

First, a contribution of this study is to test existing HC theories on a more complex setup under a vertically separated CO_2 forcing. We found the axisymmetric scaling to only be marginally useful to explain the HC response in our experiments or under the conditions present in Earth's atmosphere. It rather seems to be closer to the eddy-driven limit, which has

[1] They average contributions over both hemispheres and over latitude belts (15 to 25 °).

shown to be much better at explaining the HC strength. The energy budget decomposition framework has proven to be very widely useful and can be easily applied to all different setups.

Our results suggest that there is a sensitivity of the HC to the ratio of tropospheric warming versus stratospheric cooling. This can be seen by the opposing GMS and eddy contributions, which depend on the vertical and meridional temperature structure. Additionally, the fact that the increase of CO_2 in the stratosphere has an effect on the HC, suggests an influence of the stratospheric response on the HC. The OHU contribution cannot explain the weakening in *4xCO2 Strato*, because it is small there.

Lastly, our results show the importance of the land distribution to determine the energy input into the atmosphere and thus driving the global circulation response.

5.5. Outlook

To better isolate the effects of tropospheric and stratospheric CO_2 increase, a complementary design is to run these experiments in an aquaplanet setup. This guarantees hemispheric symmetry and lets us explore whether the opposing HC response survive without sea ice and continents.

The vertical decomposition of CO_2 concentration is quite a unique experiment. Further work could examine the radiative response, the response of the Walker circulation, mid-latitude storm tracks, or the individual feedbacks using radiative kernels. It might also be valuable to perform a more seasonal analysis and see how the HC changes in different regimes. Different vertical separations, such as fixing the tropopause height not to a climatology tropopause but using a single pressure level to avoid cross contamination or introducing intermediate layers in the vertical (upper troposphere CO_2 increase only), could strengthen the confidence in our results.

The changes in HC extent have received less attention in this study. Not only are its changes more linear than the HC strength response but also the magnitude is more robust across the hemisphere. It is surprising that the change in HC extent is related to tropospheric and stratospheric CO_2 increase. It might be interesting to further investigate the change in baroclinicity here.

5.6. Conclusion

At the core of this study lies the idea to decompose the CO_2 forcing in the vertical. In our model setup, this forcing structure has the implication that the Hadley circulation is weakening with stratospheric CO_2 and strengthening with tropospheric CO_2. This gives the opportunity to test existing HC theories against two very much different Hadley responses. Both angular momentum budget frameworks, eddies and axisymmetric,rely strongly on idealisations to isolate the underlying physical processes. The energy budget framework provided a more complete picture. Despite its shortcomings, it shows us the large qualitative differences between northern and southern hemisphere, the heat uptake at the surface due to sea ice, and how the land-sea distribution shapes the meridional distribution of net energy input into the atmosphere. Hadley cell changes can only be understood as the sum of contributions from radiative forcing, feedbacks, gross moist stability, eddies, and ocean heat uptake. These

results suggest that the stratospheric response to CO_2 influences the tropical circulation and should not be neglected in studies of HC changes.

Acknowledgement

First, I would like to thank my first supervisor Sarah, for her bright guidance, steady promotion and her valuable inputs. Also, I want to thank my second supervisor Moritz, for his constant support, encouragement, everlasting effort to get this work forward, and the intense discussions that we have had.

Thanks to my working group for all the feedback, comments, inspiration conversations, and nice get togethers that we have had. Thanks to scientific computing and administrative staff at MPI-M, who always had an open door for my numerous questions and requests. Thanks to the external visitors at MPI-M for their useful comments on how to advance this study and providing the bigger context. Thanks to all my previous teachers for their patience and for bringing me back to earth.

Also, I would like to thank everyone who has helped me through the difficult times that were coinciding with this work.

This work would not have been possible without the constant care of my family, and my beloved friends, who always kept me motivated.

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